

Hybrid-INoT: internal role segregation and external verification for code generation

Daniil Privezentsev^{1*}

^{1*}National Research University Higher School of Economics, Moscow, Russia.

Corresponding author(s). E-mail(s): daprivezentsev@edu.hse.ru;

Abstract

Hybrid-INoT combines planning, implementation and critique within one model response with external executable verification. We formalize its single-pass core and test role-labelled versus neutral instructions crossed with one versus three model calls, holding the operations and full task context fixed. A direct solver is the fifth condition. In the largest completed study, GPT-5.6 Luna with medium reasoning receives 15,000 assignments on 1,000 BigCodeBench task clusters with three fresh repeats per condition: 200 task IDs are reused from the initial factorial study and 800 are additional IDs. All 14,961 complete candidates undergo native evaluation; 985 tasks pass reference controls. Observed success ranges from 50.07% to 54.59%. Role-minus-neutral quality differences are **−0.75** percentage points within one call and **+1.35** within three calls; neither comparison is statistically significant after Holm adjustment across the four prespecified tests. Three-call execution lowers quality by 4.17 points for neutral instructions and 2.13 points for role-labelled instructions, while requiring 2.67 and 2.78 times their single-call API-equivalent valuations. Whole-task and source-group intervals, repeated-outcome tables and explicit missingness support these comparisons; the primary quality contrasts use complete-pair task estimators and retain 264 unknown endpoints. The earlier 200-task factorial and 80-task label-by-layout studies retain their separate model and sampling scopes. Together, the findings identify a cost of the tested call structure without establishing a general quality benefit from role names, quality non-inferiority, or independent internal agents.

Keywords: Hybrid-INoT, internal role segregation, external verification, token efficiency, factorial design, code generation

1 Introduction

A software engineering agent must understand a specification, use the relevant code and documentation, construct a solution, and test it. These functions can be distributed among a planner, worker and critic. If separate model calls receive the same material, the context must be transmitted repeatedly. We investigate whether these functions can be retained with fewer calls while leaving correctness checks to external tests.

Hybrid-INoT addresses this question by combining internal role organization with an external verification layer. Planning, implementation and critique are prescribed in a shared model call; an external process evaluates the resulting program. The design draws on INoT’s use of internal dialogue [5], but the role-labelled configuration examined here is distinct from the original PromptCode algorithm. It separates program generation from correctness checking. Long shared context motivates the resource model; it does not by itself guarantee that a single call will preserve quality.

The original Hybrid-INoT implementation also included context selection and selective reruns. The preliminary comparison changed context selection, rerun policy, and call topology together, so it cannot identify the contribution of internal roles. The present evaluation isolates the architecture’s single-pass core: full supplied context, one final candidate, prescribed planning/implementation/review and subsequent native testing. Compression and verification-driven retries are disabled. Evaluating the components separately tests these architectural choices within the original engineering problem.

The core has two distinct empirical questions. Does assigning role names add value when the stage operations stay fixed? Does performing the operations in one response improve the quality–resource tradeoff relative to distributing them across three calls? A direct solver tests whether the staged organization is useful at all. A separate INoT adaptation examines the related algorithmic alternative. Each condition receives the same supplied task and context and undergoes the same subsequent evaluation. The design never treats a model’s self-critique as the measured verification result.

The contributions are:

1. An operational account of the Hybrid-INoT single-pass core, its state and its external evaluation boundary, with a direct mapping to every implemented control.
2. A conditional shared-context token balance explaining when fewer calls can reduce resources, without assuming equal output lengths or asserting a measured long-context crossover.
3. A compression-free 2×2 factorial study with a direct comparator on 1,000 BigCodeBench tasks, three repeats per condition and 14,961 native-checked programs. Four prespecified task-level quality contrasts separate role labels from call structure.
4. Complementary evidence from the initial 200-task factorial study, an 80-task label-by-layout experiment, development programs, an INoT adaptation and a one-issue software-repair demonstration. Complete repeat outcomes and instructions are retained in the appendices.

Role ablation and interaction-count studies have close predecessors. The contribution here is the specified architectural evaluation and its auditable evidence, not the invention of role collaboration. The larger experiment finds higher resource use and lower test success when the stages are split across three calls. It does not establish a general quality advantage of role names or quality non-inferiority.

2 Related work

2.1 Closest methodological antecedent

Self-collaboration Code Generation via ChatGPT [1] already ablates role instructions and maximum interactions in Sections 5.2–5.3. Removing roles or varying interaction is therefore not a novelty claim of the present study. Table 1 distinguishes the implemented interventions.

Table 1 Method comparison with the closest antecedent [1], Tables 3–4 and Appendix B.1. Interaction rounds and fresh model calls are different quantities.

Aspect	Self-collaboration paper	Component experiment
Labels	Role versus role-excised instructions	Identical operation sentences; only the stage prefixes change
Interaction	Feedback loop; maximum interaction varied	One or three fresh turns; fixed sequence, full exposed history
Design	Separate role and interaction ablations	Joint label-by-call matrix on every assigned task
Tasks	HumanEval, MBPP and extended tests	Reserved BigCodeBench sample; pre-generation reference and incorrect controls

The joint design estimates the interaction of role labels and call topology under a fixed generation interface. The evaluated programs, native outcomes and token components make its quality and resource comparisons inspectable. This extends the empirical comparison without claiming a new principle of role collaboration.

Self-collaboration is the closest external method because its role ablation addresses the same attribution question. Our external comparison uses the pinned 2024 author implementation, whose generated-test execution differs from the simulated testing described in the 2023 paper (Section 14.3). The frozen comparison assigns SCC, fresh SR and fresh SN to the same 1,000 tasks with three repetitions; it contains no role-excised SCC arm. It therefore estimates differences between complete workflows, including feedback and stopping, while SR–SN and MR–MN isolate the specified labels within our component design. Two development checks establish execution feasibility, but the main comparison remains unfinished. D provides the minimal comparator. INoT* addresses a separate internal-dialogue algorithm question.

2.2 Internal dialogue and repository methods

INoT describes compact PromptCode instructions for internal dialogue and reasoning [5]. This motivates evaluating a disclosed algorithm adaptation, while a separate minimal label intervention addresses whether named roles themselves change results. The two comparisons answer different questions: the INoT instruction changes the reasoning procedure, whereas the factorial label contrast preserves the operation sentences. The original Hybrid-INoT comparison additionally changed compression and reruns; its results are described in the historical appendix and are not pooled with the new experiment.

Single- versus multi-agent work under matched thinking-token budgets emphasizes that additional calls and additional computation are distinct interventions [6]. Here, actual completion length is not held to a matched hard budget. We therefore report token components and list-price valuations alongside quality. These measurements describe the implemented protocols; they do not identify an optimal allocation at a fixed inference budget.

Agentless uses a structured localization–repair–validation workflow [7]; SWE-agent couples the model to a repository interface [8]. Both add retrieval, tools, and environment interaction beyond fixed-context function generation, so their published scores are not directly comparable with this paired matrix. BigCodeBench [9] supplies diverse library-dependent tasks with executable tests, whereas SWE-bench [10, 11] evaluates repository changes against issue-specific tests. Native evaluation strengthens outcome validity, but passing a reference control does not resolve every disagreement between task prose and tests.

Feedback is a separate design choice from role naming. Self-Refine alternates model-generated feedback and revision using the same model [3]. Reflexion retains verbal reflections between trials; its programming experiments use self-generated executable tests as feedback [4]. Our component conditions receive no execution feedback during generation and follow a fixed sequence, so they do not reproduce either adaptive procedure. The SCC author-code comparison adds generated-test feedback within its complete workflow. In our experiments, benchmark tests and reference programs remain outside the generation interface; internal SCC execution never determines benchmark success. These distinctions make feedback access and stopping part of the external method comparison, rather than evidence for a role-label effect. Self-Refine and Reflexion provide methodological context; neither has been run as an experimental arm here.

Persona studies motivate a narrower claim than “roles improve reasoning.” Zheng et al. find that adding system personas does not reliably improve their factual-question evaluations [16]. Luz de Araujo et al. distinguish performance, robustness to irrelevant persona attributes, and fidelity to the assigned expertise; their results vary across models and tasks [17]. These are not tests of our coding prompts. They motivate changing labels while retaining the requested operations, and reporting the result for the exact labels used rather than for all possible personas.

Multi-agent evidence also depends on its comparator. Choi et al. attribute much of the gain in their NLP debate experiments to independent voting and show why communication must be separated from aggregation [18]. Conversely, Wunderlich et

al. report configurations where debate or mixture-of-agents improve the compute-accuracy tradeoff relative to self-consistency on MMLU-Pro and BBH [19]. Together these findings support conditional evaluation, not a universal claim for or against multiple agents. Our neutral serial pipeline is not voting, a mixture of independent models, or a matched-budget ensemble. This study provides a within-model factorial test on executable code, preserving traces and accounting for available resource counters; it does not estimate general superiority over those alternatives.

3 Formal problem and the Hybrid-INoT core

3.1 Engineering task and observable success

An engineering task is represented by $(x, \mathcal{C}_0, \mathcal{V})$: a specification x , a fixed context supplied with the task $\mathcal{C}_0$, and an external verification procedure $\mathcal{V}$. The output consists of a candidate program y and its verification record. Code generation and verification are different operations: the model proposes a solution; original executable tests determine the benchmark outcome in a controlled environment. A textual claim that the solution is correct cannot replace those tests. For repository tasks, $\mathcal{C}_0$ denotes the disclosed retrieved packet, not an assertion that the full repository was supplied.

For an execution protocol a , the engineering objective is to increase the probability of verified success while controlling expected token use and generation cost. We therefore report quality and resources separately rather than optimize an unvalidated weighted score. Observed programs that fail tests are failures; unavailable generation or evaluation cannot be turned into a verified success or a fabricated test failure. Section 5 specifies the common eligibility controls and statistical endpoint.

3.2 Internal role organization and external verification

Hybrid-INoT uses a division of responsibility: planning, implementation and critique are prescribed within a model prompt, while verification is performed outside the model. Here, *hybrid* denotes this combination of internal instructions and external executable evidence. It does not require a learned deployment router. The original planner-worker-critic terminology corresponds to the executed labels **planner**, **implementer** and **reviewer**. Internal role segregation here denotes separation of instructed functions, not independently observed latent agents.

The single-call core studied here prescribes three operations: analyze requirements and edge cases, implement using the plan, then review and revise against the requirements. A single model call receives all three instructions and the complete task context and returns one final program. The experiment does not observe hidden plans or critic scores. The three operations are therefore instructed functions, not three measured internal subroutines. The precise prompts, including the final-output contract, appear in the appendix.

In the primary experiment, SR is this role-labelled single-call condition. SN is its role-label ablation with identical operation sentences. MR and MN distribute the corresponding operations across three fresh calls, and D removes the staged procedure altogether. All five receive the same subsequent native evaluation. The design tests

whether the proposed internal organization contributes beyond a plain instruction and what changes when the same operations are distributed across call boundaries. Hybrid-INoT has no privileged access to verification.

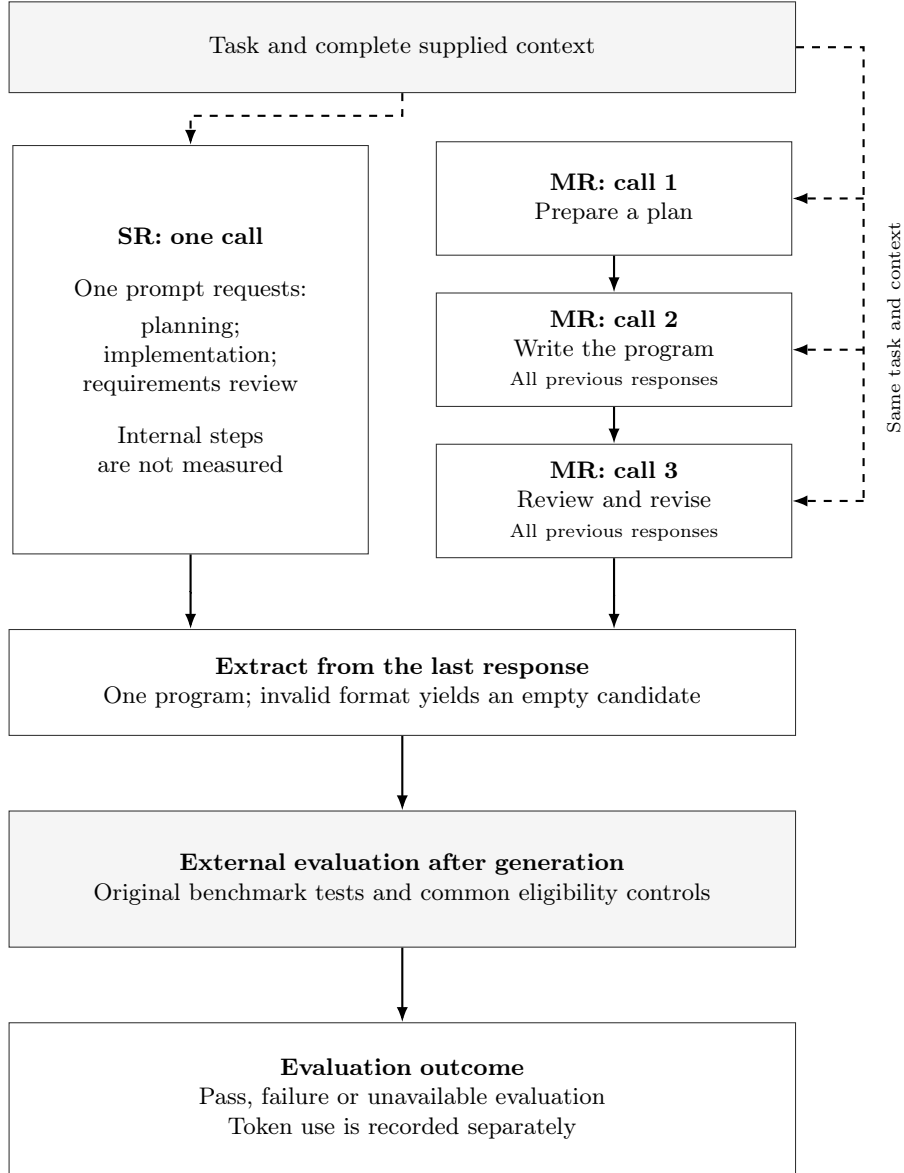


Figure 1 Two generation configurations: SR requests three functions in one prompt; MR distributes them across three calls. Arrows between MR calls carry all complete previous responses; each call also receives the task and context. The figure shows the path of an observed response. Incomplete generation is handled separately in Algorithm 1. All five conditions in Table 2 share external evaluation, with no test feedback to the model.

3.3 State, execution and stopping

The observable execution state is $\sigma_j = (x, \mathcal{C}_0, a, j, \mathcal{H}_j)$, where a fixes the assigned condition, j is the call index and $\mathcal{H}_j$ contains all complete preceding exposed responses. The model configuration is fixed within the comparison. D, SR and SN use $n_a = 1$; MR and MN use $n_a = 3$. In a multi-call condition, $\mathcal{H}_j$ is forwarded in full. The supplied context is unchanged: $\hat{\mathcal{C}} = \mathcal{C}_0$. Condition assignment is fixed before generation, so it is not an outcome-dependent routing policy.

Algorithm 1 Program generation and outcome evaluation

Input: task, complete supplied context and preassigned condition.

Output: observed program, evaluation outcome and known resource use.

I. Generation

1. Determine the number of calls: one for D, SN or SR; three for MN or MR.
2. Execute the calls sequentially. Give each call the task, unchanged context and assigned instruction. Give calls two and three every complete previous response as well.
3. Retain each call’s prompt, complete response, terminal status and token counters.
4. If a call is incomplete or resource accounting cannot be verified, retain the available evidence and stop that assignment as unavailable. Do not retry it automatically.
5. Extract one program from the last completed response. Retain an empty candidate if the format is invalid.

II. Evaluation after generation ends

6. Evaluate every observed candidate with the original benchmark tests.
 7. Match the evaluated program to its native report and common eligibility controls.
 8. Record pass, failure or an unavailable outcome. Retain all known resource counters; do not return test results to the model.
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Algorithm 1 describes the generation and evaluation stages of the experiment; the native harness is a subsequent evaluator, not a tool invoked inside the model call. The call deadline and the fixed call count bound the generation procedure. This is a stopping rule, not a theorem of convergence or correctness. A rejected completion never triggers an automatic second attempt for the same assignment. The disclosed continuation concerns only assignments for which no call had started.

This configuration fixes the original architecture’s optional controls: context selection is the identity, the mode is assigned, the number of final candidates is one, and verification-driven reruns are disabled. These choices isolate the internal-role question from compression, a small checking model, and adaptive retries. They also define the scope of the evidence: the experiment evaluates this core, rather than every possible Hybrid-INoT deployment.

4 Analytical model of shared-context costs

4.1 A conditional token balance

The architectural motivation is repeated transmission of common material. Consider an additive accounting model in which the common task/context block contributes S tokens each time it is supplied. An internal configuration H transmits it once; an external configuration M transmits it in each of m calls. Let J_H, J_M denote all other input tokens, including instruction wrappers and forwarded responses, and let O_H, O_M denote all output tokens. Define $R_H = J_H + O_H$ and $R_M = J_M + O_M$. Then

$$T_H = S + R_H, \quad T_M = mS + R_M, \quad T_M - T_H = (m - 1)S + R_M - R_H. \quad (1)$$

This is an accounting identity under the stated block model. It does not assume equal output length. In particular, longer internal reasoning can offset reduced context transmission; external intermediate responses contribute both when generated and when supplied again.

Proposition 1 (Conditional sufficient resource advantage). *For $m > 1$, suppose $\mathbb{E}[R_H - R_M \mid S] \leq K$ throughout a specified context range, with a constant $K \geq 0$ independent of S in that range. Under Eq. (1),*

$$\mathbb{E}[T_M - T_H \mid S] \geq (m - 1)S - K.$$

Consequently, the internal configuration has a positive expected token advantage wherever $S > K/(m - 1)$ within that range.

Proof Subtract the two balances in Eq. (1), condition on S , and apply the assumed upper bound on $\mathbb{E}[R_H - R_M \mid S]$. $\square$

The residual bound is an assumption, not a measured property of the model. If output or coordination costs change with context so that the bound fails, the inequality gives no crossover. The present sample does not manipulate context length or estimate K ; it cannot identify a long-context crossover or routing threshold. The earlier compressed pilots do not identify either quantity.

The empirical ledger uses actual provider counters,

$$T_a = \sum_{j=1}^{n_a} (I_{a,j} + O_{a,j}), \quad (2)$$

rather than estimating totals from Eq. (1). Unknown usage remains unknown. The conceptual block S is not equated with a measured additive partition of provider input: tokenizer boundaries and client overhead can prevent such a partition. This distinction avoids presenting a theoretical balance as an observed decomposition. At fixed call topology, the shared term cancels between role and neutral conditions; any resource

difference then belongs to their wording-induced and other residual differences. That is the reason to include both label controls.

4.2 Quality and money remain separate constraints

Write $Q_a = \mathbb{E}[Y_t(a)]$ for verified success and $\bar{T}_a = \mathbb{E}[T_a]$ for expected tokens. The original engineering notion of token utility can be expressed as $U_a = Q_a/\bar{T}_a$, using a ratio of expectations on a common, fully observed population. A larger U_a can coexist with worse Q_a ; it is not a quality-preservation certificate. Because the observed quality and resource subsets also differ in this study, the primary report keeps paired quality differences and paired resource ratios separate, without constructing a composite utility ranking or assigning unvalidated maintainability weights.

The monetary counterpart must use the tariff schedule for each input, cached-input and output component. The API-equivalent quantities V and V_0 in Section 5 implement this requirement with the recorded counters. Token advantage alone does not establish monetary advantage when cache fractions and output shares differ. Neither quantity measures the subscription bill or total ownership cost; evaluator compute, engineering work and deployment operations are outside the observed model ledger.

The balance describes a possible efficiency mechanism. The factorial experiment below addresses whether quality changes, whether labels add value, and whether the observed resource difference favors the internal configuration. This explanation does not add a confirmatory hypothesis to the four tests frozen before the primary run.

5 Component evaluation: study design and estimands

5.1 A factorial intervention without compression

The main experiment evaluates the Hybrid-INoT core and its controls by crossing call topology (one or three fresh model calls) with stage labels (neutral or role-labelled). Table 2 defines the four factorial cells and the direct comparator. All structured conditions prescribe the same operations: analyze requirements and edge cases, implement a solution using the plan, then review and revise against the requirements. The label intervention replaces “stage 1”, “stage 2”, “stage 3” with “planner”, “implementer”, “reviewer”; the operation sentences remain unchanged. In a single turn, all three instructions precede the task. In three turns, each turn receives its stage instruction, the full task, the full supplied context, and all complete previously exposed responses. The final-output contract is identical: exactly one fenced Python program. The direct comparator receives only the implementation instruction and the same output contract.

Compression is absent in every cell. This replaces the earlier comparison of three uncompressed calls with one compressed call; it does not estimate the effect of the historical compressor. The isolated label contrast is a prompt intervention, not evidence of independent internal agents. The topology contrast includes intermediate information flow and repeated inference, rather than a purely mechanical call-count effect. The untested router and auxiliary checker are removed from the claimed method.

Table 2 Implemented treatment matrix. The topology factor includes the exposure of intermediate responses across fresh calls.

Code	Calls	Stage labels	Operations
D	1	None	Direct implementation
SN	1	Neutral	Plan, implement, review
SR	1	Roles	Plan, implement, review
MN	3	Neutral	Plan, implement, review
MR	3	Roles	Plan, implement, review

Let $Y_t(a)$ indicate that the single final program for task t in condition a passes the original benchmark tests in the validated environment. The four predeclared paired quality effects are

$$\Delta_{R,1} = \mathbb{E}_t[Y_t(\text{SR}) - Y_t(\text{SN})], \quad \Delta_{R,3} = \mathbb{E}_t[Y_t(\text{MR}) - Y_t(\text{MN})], \quad (3)$$

$$\Delta_{C,N} = \mathbb{E}_t[Y_t(\text{MN}) - Y_t(\text{SN})], \quad \Delta_{C,R} = \mathbb{E}_t[Y_t(\text{MR}) - Y_t(\text{SR})]. \quad (4)$$

Each null is a zero quality difference. The descriptive interaction is $\Delta_{R,3} - \Delta_{R,1}$. Expectations describe the sampled task population under the implemented protocol and its model sampling, not a provider-independent property. Available-case estimates require complete paired outcomes; with missingness they describe that observed subset unless stronger assumptions are made.

5.2 Task allocation and a separate development stage

The task source is BigCodeBench v0.1.4 [9]. The main allocation contains 200 new task IDs drawn from a random permutation of the reserved pool, using Python’s pseudorandom generator with seed 20260908 after lexicographic sorting of task IDs. The first 200 eligible IDs in that random order are taken after excluding eight dataset-card examples (/0–/7) incidentally exposed during license verification. This is neither a first-row convenience sample nor a difficulty-selected subset. Task IDs, model inputs and evaluator data are fixed before control testing; no failed control causes replacement. The 40 development tasks are disjoint from these 200. Development uses two labelled repeats and 400 candidates; the main allocation uses one new repeat label and 1,000 candidates. The 1,000 assignments therefore represent at most 200 independent task units, not 1,000 independent tasks. Related benchmark tasks may further reduce effective independence. Public availability leaves training contamination unresolved.

An assignment denotes a planned generation for a specified task, condition and repetition; once started, it counts as an attempt. All five conditions are assigned to every main task. Four disjoint index-modulo-four shards each contain 50 tasks. Within each shard, a frozen pseudorandom order determines task blocks and condition order; two workers process each shard, with at most eight simultaneous experimental CLI processes. The allocation and implementation are fixed before dispatch. Appendix B

gives input construction and evaluation settings. The model does not receive benchmark tests, reference solutions or test feedback during generation. Repeated labels identify fresh protocol observations; they are not provider sampling seeds.

5.3 Model execution and complete exposed traces

Generation uses the requested `gpt-5.4-mini` alias, medium reasoning, and official Codex CLI 0.153.4 through ChatGPT subscription authentication [12, 15]. The published model tariff makes resource valuation auditable; model choice is an economical coding configuration available for this study, not a demonstrated global price minimum. The CLI uses a clean configuration without user or project instructions. Tools, browsing, delegation and test execution are disabled. Every call creates a fresh ephemeral thread in an empty working directory. The accepted event schema requires exactly one completed response and a valid usage record; an unexpected tool call, retry, or compaction event invalidates the fixed text-only treatment.

Every submitted call retains its exact prompt, arguments, event stream, standard error and terminal status. Completed records also retain the final text and token components. Each multi-call prompt is checked to contain every preceding exposed response in full. No hidden provider reasoning is claimed to be available. The model alias does not fix immutable weights, and no verified temperature or model seed is exposed. A 180-second deadline applies to each turn. An input above 65,536 UTF-8 bytes is rejected rather than shortened. There is no matched hard output-token allowance; actual output length and resource consumption are outcomes of the treatment.

The original execution rule stops a shard after its first failed cell. Transport/deadline failures therefore leave unrelated assignments unsubmitted. A separate amendment to the run protocol, recorded before held-out quality evaluation or inspection, permits only those never-submitted cells to continue. Any cell with even one started stage is permanently excluded from continuation. The pending list and first-attempt records are fixed before continuation; prompts, settings, deadlines and original within-shard order are preserved. An explicit transport error or deadline ends a continuation cell without retry; quota, authentication or unclassified errors stop further dispatch. First attempts, continuations and missing outcomes remain distinguishable. Thus analysis follows predeclared contrasts under an amended execution protocol, not an unchanged preregistered run.

5.4 Native evaluation and eligibility controls

The unchanged BigCodeBench evaluator runs the v0.1.4 task set in `instruct full` mode. Each native run uses an offline, non-root, read-only container with a 3-GB memory cap, two CPUs and two evaluator workers. Evaluation settings and the dependencies added during reference-control checks are given in Appendix B. Strict extraction selects the sole final Python block; an invalid final format yields an empty observed program and a format-failure flag, without alternative-answer selection or repair.

Before model generation, original gold programs and deliberately incorrect programs are tested for all 200 tasks. Dependency amendments are confined to this

control phase, retain earlier failed attempts, and trigger fresh full-allocation controls. The final environment passes 193 gold controls and rejects all 200 incorrect controls. Seven tasks are consequently unavailable for quality measurement in every condition; their assignments and generation resources remain counted. The final image adds pinned dependencies and offline NLTK resources to the development environment. Native gold diagnostics identify DNS failures (/101, /590), missing TensorFlow (/418), an unsupported encoder argument (/686), degenerate-moment assertions (/276), and archive/logging assertions (/14, /1028). A later read-only inventory confirms that the same image lacks the external commands used by the latter two tasks; attributing their assertions to those missing commands is a source-based diagnostic inference. No subsequent diagnosis changes the frozen eligibility gate. These checks establish evaluability and discrimination against deliberately incorrect programs, not complete prompt–test semantic validity.

Only actual observed candidate programs are submitted for evaluation. Each native result is matched to the evaluated program and task in the validated environment; incomplete evaluator runs or inconsistent reports cannot supply quality observations. The raw native status is retained alongside a separate eligibility-gated analysis status. Missing generations are not fabricated native failures. The same eligibility gate applies to all conditions and the exploratory INoT replication.

5.5 Inference, missingness and resource valuation

For each of the four planned quality contrasts, the analysis reports complete-pair counts, both discordant counts, paired mean difference, exact two-sided McNemar binomial test and Holm-adjusted p across the four tests at familywise $\alpha = 0.05$. These four McNemar tests alone define the primary rejection family; neither bootstrap intervals nor sign-flip sensitivity tests determine a primary rejection. Percentile intervals use 10,000 task bootstrap resamples with analysis seed 20260908; these descriptive intervals are not simultaneous intervals. The interaction, resource contrasts and direct-solver comparisons remain descriptive. The interaction uses tasks with complete outcomes across all four factorial conditions; it need not equal the difference between two separately estimated label contrasts if their available-task subsets differ. Development repeats are averaged within task for its separate descriptive analysis and never pooled into the main tests.

Two hundred independent evaluable tasks would yield a worst-case single-proportion normal 95% half-width of approximately 6.9 percentage points. This is a precision rationale, not an 80%-power calculation for a paired effect; paired uncertainty depends on discordance. No application-specific acceptable quality loss has been externally justified, so no non-inferiority margin or quality-preservation decision is used. A margin selected solely to fit statistical precision would not establish an acceptable practical loss. Per-condition quality uses evaluable observations and reports all-assignment missingness ranges: unavailable outcomes are first all failures, then all successes. Those ranges are neither confidence intervals nor missing-at-random corrections.

Completed CLI turns expose input I , cached-input C and output O counters. Cached input is a subset of input; any separately exposed reasoning count is a subset

of output. Neither is counted twice. Published standard API rates are USD 0.75, 0.075 and 4.50 per million tokens, respectively [14]. We report

$$V = \frac{0.75(I - C) + 0.075C + 4.50O}{10^6}, \quad V_0 = \frac{0.75I + 4.50O}{10^6} \quad \text{USD.} \quad (5)$$

V is a counterfactual API list-price valuation of measured subscription-channel usage, not a payment or invoice; V_0 is a no-cache sensitivity. Both exclude research-assistant work, software development, environment construction and evaluator compute. Resource use for each candidate is calculated by summing its completed calls; the reported means are then calculated across candidates. A separate inventory includes observed usage from interrupted candidates and explicitly counts turns without final usage; unknown usage is never zero-imputed. Resource ratios use the same complete paired task subset in numerator and denominator. A lower observed valuation means the ratio of paired mean valuations is below one, equivalently a negative mean paired difference. This descriptive criterion is distinct from monetary superiority: no confirmatory economic threshold or joint quality–cost decision rule was registered, and the resource intervals do not establish such a claim. Timing and cache state can influence valuation, especially across separately dispatched batches.

5.6 Scope of the completed Luna extension and amended SCC study

The preceding design, estimands and McNemar tests describe the initial 200-task component study and remain its historical analysis. The completed extension is a separate 1,000-task Luna study: 15,000 assignments, 14,961 complete candidates, 985 control-eligible tasks, and 264 unknown quality assignments. It reuses the initial 200 task IDs and adds 800 IDs from the fixed randomized order, so it is not a new held-out sample. Its original three-repeat task estimator and four-test Holm family are retained; it is not pooled with the initial mini study. For this Luna extension, the counterfactual API-equivalent valuation uses the verified current rates of USD 0.20 per million input tokens, USD 0.02 per million cached-input tokens, and USD 1.20 per million output tokens. This rate statement is separate from the initial mini-study tariff above and is neither a subscription invoice nor a claim about total cost. The declared model alias and repeat requests do not fix immutable provider weights or certify independent generations.

The SCC comparison follows the amended protocol in its own scope. The selected prefix is the first 2,000 task-repeat blocks of the original 3,000-block schedule: 6,000 method assignments over 956 tasks, with 941 control-eligible tasks. The remaining 3,000 assignments are administrative nonselection. The amended primary estimand pairs the same selected repeat labels only when both SCC and the comparator have observed native endpoints, averages the paired binary differences within task, and weights contributing tasks equally. SCC–SR and SCC–SN form the fixed two-test Holm family; percentile intervals use 10,000 whole-task draws, seed 20260911, NumPy PCG64 and linear interpolation. Generation is ongoing and native outcomes have not been inspected. Eight interrupted submitted attempts remain paused unknowns and

are not retried. Resource analyses require complete priced workflows and known counters for every submitted turn; partial subtotals remain separate. Resource accounting for the Luna extension follows the definitions and complete-task rules in Appendix B; the Luna rates are stated above. Interrupted workflows and unknown usage remain reported separately and are never zero-imputed.

6 Exploratory INoT algorithm replication

The four factorial conditions are component interventions, not a replication of a published method. We therefore add a separately frozen prompt-level replication of INoT [5], labelled INoT*. It uses the same 200 tasks, the full supplied context, GPT-5.4 mini medium, the final-output contract and the native control gate. Each task receives one fresh call using two workers under the same 180-second deadline. The independently worded PromptCode instruction describes two virtual debaters, initial answers, mutual arguments, critiques, rebuttals, updates and an agreement check, with a ten-round cap. This is conceptual guidance read by the model, not code executed by the host or a verified sequence of hidden agent interactions.

The stopping rule follows the source paper’s prose: agreement or the round cap. The corresponding source-code listing has an ambiguous loop condition and leaves the final answer unspecified if agreement is never reached. This replication explicitly returns the latest answer from virtual A in that case. Image augmentation is omitted for these text-only tasks. The exact independently worded instruction appears in the prompt appendix. No verified executable author implementation was identified for this setup, and no author validation of the adaptation is claimed.

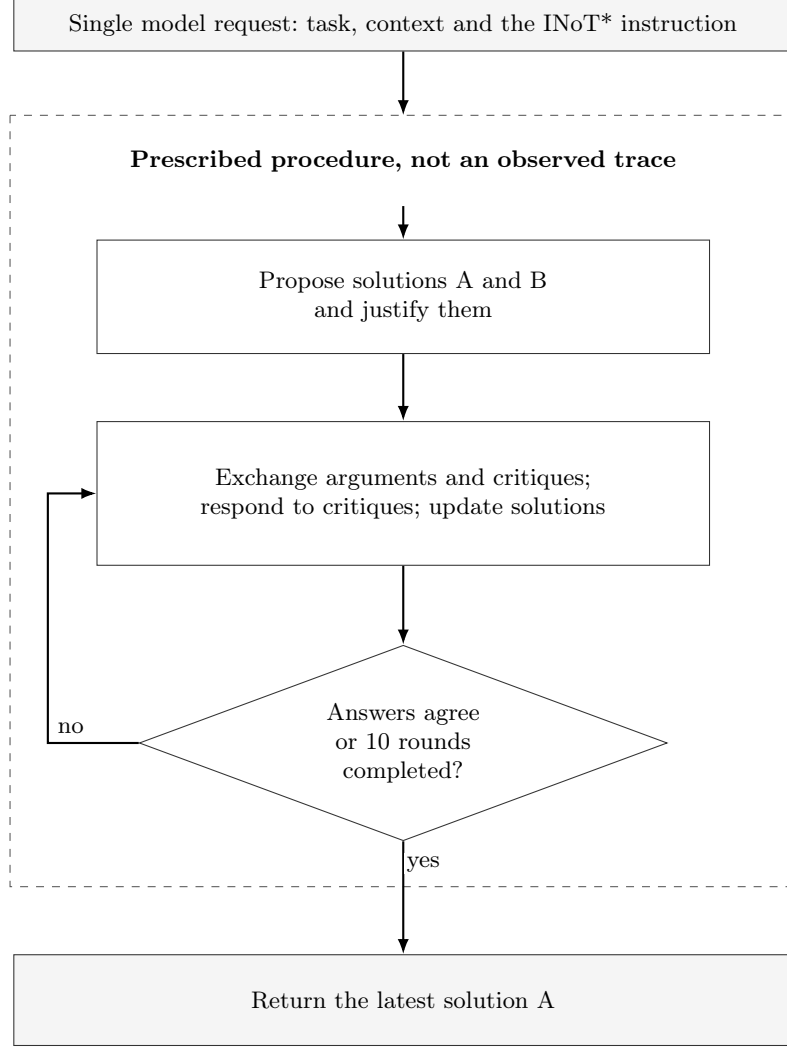


Figure 2 How to read the INoT* instruction. The dashed region describes operations prescribed in one prompt, not an executed host loop or measured internal agents. Under agreement, A and B have the same answer; without agreement after ten rounds, the instruction selects the latest answer from A. Appendix A.1 gives the literal wording.

An independent administrative continuation follows the same never-submitted-only rule: the initial timed-out assignment is not rerun, and the remaining 137 untouched assignments are frozen and submitted once. All first-attempt and continuation traces remain separate. Comparisons of INoT* with D and SR are exploratory and outside the four tests in the initial factorial study. Task pairing controls membership, but the separate batch retains timing, cache-state and possible provider-drift differences. An observed result is evidence about this disclosed adaptation, not an exact reproduction of the original paper’s reported scores.

7 Repeated factorial study on 1,000 tasks

7.1 Design and completed evaluation

The larger study retains the five conditions D, SN, SR, MN and MR and the full supplied task context. It uses GPT-5.6 Luna with medium reasoning through Codex CLI 0.153.4. Each condition receives three fresh generations per task, giving 15,000 assignments across 1,000 task clusters and 27,000 planned calls. The repeat labels identify fresh requests; they do not set a provider sampling seed. Model-call deadlines are 600 seconds. Neither context compression nor repair based on benchmark test results is allowed. The task clusters are the task-level units; 264 quality endpoints remain unknown, and the four contrasts use complete-pair task estimators with 982, 964, 968 and 971 task pairs, respectively.

The allocation contains the 200 tasks used in the initial factorial study and 800 additional tasks from the original randomized order, excluding development tasks and the separate layout study. All programs are generated afresh; earlier answers are not pooled with them. Thus this is a larger experiment with a different requested model, not 1,000 newly held-out tasks or a replication with immutable model weights. Reference and incorrect-program controls, completed before generation, make 985 tasks eligible for quality measurement. Task similarity and cross-study overlap are described separately below.

Generation produced 14,961 complete candidates; 39 submitted assignments remained incomplete. Every complete candidate underwent the native BigCodeBench evaluation. The analysis retains all 15,000 assigned rows, including 264 unknown quality endpoints arising from generation or control availability. Unknown endpoints remain distinct from test failures. Table 3 separates generation completion from quality observability; three attempts at one task are not treated as three independent task units.

Table 3 Luna factorial study: assignment and observation counts.

Condition	Assigned	Generated	Observed	Passes	Success (%)
D	3000	2996	2951	1611	54.59
SN	3000	2998	2953	1604	54.32
SR	3000	2997	2952	1581	53.56
MN	3000	2983	2938	1471	50.07
MR	3000	2987	2942	1514	51.46

Observed means eligible native quality endpoints, not distinct tasks. Success divides passes by observed repeat outcomes. Each condition has three assigned repeats per task.

For each of four prespecified contrasts, both conditions must have all three quality endpoints available. The difference between their three-repeat means is calculated within each task, and tasks receive equal weight. The four contrasts therefore use their

respective complete-pair task counts. The primary two-sided t -tests form one four-test Holm family at $\alpha = 0.05$. The 95% task-bootstrap intervals are descriptive, with 10,000 resamples and seed 20260909; they do not replace the multiplicity correction. The experiment estimates per-generation test success, not success after choosing the best of three candidates. Appendix B.5 gives the calculations and missingness rules.

7.2 Roles, call structure and quality

Table 4 Luna factorial quality contrasts. Differences and bounds are percentage points.

Contrast	Tasks	Difference	95% interval	Holm p	Assigned bounds
SR – SN	982	−0.75	[−1.97, 0.44]	0.2357	[−2.33, 0.83]
MR – MN	964	+1.35	[0.17, 2.56]	0.0599	[−0.63, 3.37]
MN – SN	968	−4.17	[−5.65, −2.75]	1.34×10^{-7}	[−6.00, −2.37]
MR – SR	971	−2.13	[−3.43, −0.82]	0.0044	[−3.83, −0.30]

Tasks require three observed repeats in each compared condition. Assigned bounds use all 1,000 tasks and every assigned repeat; they are not confidence intervals.

Within one call, the role-minus-neutral difference is −0.75 percentage points, with a 95% interval of [−1.97, 0.44]. Within three calls it is +1.35 points, with interval [0.17, 2.56]. The latter interval excludes zero, but its Holm-adjusted $p = 0.0599$ does not meet the specified familywise threshold. Neither role-label comparison therefore supports a rejection in the primary family.

Splitting the stages across three calls reduces observed test success relative to keeping them within one response: by 4.17 points for neutral instructions and 2.13 points for role-labelled instructions. Both comparisons reject after Holm adjustment. Their full-assignment identification bounds also remain below zero (Table 4); this particular direction does not depend on assigning favourable values to missing binary outcomes. These bounds address missingness, not sampling uncertainty or model drift.

The descriptive role-by-call-count interaction is +2.08 points, with interval [0.42, 3.74], on the common 963-task subset. This describes how the two role contrasts differ on the same tasks; it is outside the four-test family. None of the four primary task-difference vectors is degenerate. The design includes no justified non-inferiority margin or joint quality-cost decision, so the intervals do not establish quality preservation.

7.3 Resource use and robustness

Single-call roles have a valuation ratio of 0.939 relative to neutral stages, or 6.08% lower API-equivalent valuation on 997 complete three-repeat task pairs. The token ratio is 0.986. Within three calls, the corresponding valuation ratio is 0.978 and the token ratio 0.990 on 979 task pairs. These resource differences are descriptive; their magnitude is smaller than the initial mini study’s single-call difference.

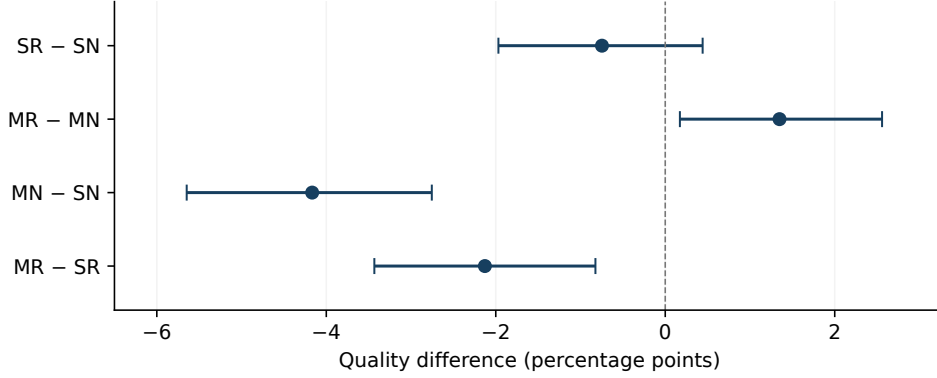


Figure 3 Luna factorial study: task-averaged quality differences and descriptive 95% bootstrap intervals. Positive values favour the first condition. Primary decisions use the separate four-test Holm family in Table 4.

Table 5 Luna factorial resource ratios: first condition divided by second.

Contrast	Tasks	Tokens	Valuation	95% interval	No cache
SR - SN	997	0.986	0.939	[0.93, 0.95]	0.955
MR - MN	979	0.990	0.978	[0.97, 0.99]	0.986
MN - SN	983	3.078	2.672	[2.63, 2.71]	2.777
MR - SR	986	3.090	2.780	[2.74, 2.82]	2.867

Ratios divide sums of three-repeat task means on complete resource pairs. The displayed interval is for the API-equivalent valuation ratio. No cache removes its input-cache discount; subscription billing is not estimated.

The resource change associated with call structure is much larger. Three-call neutral instructions require 3.08 times the tokens and 2.67 times the valuation of single-call neutral instructions. With roles, the ratios are 3.09 and 2.78. Removing the cache discount leaves valuation ratios of 2.78 and 2.87, respectively. These values concern the implemented full-context workflows and published API-equivalent rates, not subscription charges or total research costs. Token counts include repeated inputs and exposed intermediate answers; provider reasoning is not added twice.

The supplementary source-group bootstrap preserves the task-weighted point estimates. Under the combined instruction/code similarity rule, the single-call role interval is $[-1.99, 0.47]$ points and the three-call role interval $[0.14, 2.60]$. The topology intervals remain negative: $[-5.69, -2.72]$ for neutral instructions and $[-3.44, -0.79]$ with roles. These intervals use the predefined groups and are sensitivity results, not another set of significance tests. Similarity screening cannot establish semantic independence or exclude training contamination.

The larger experiment supports a protocol-specific architectural conclusion: for these tasks and settings, transmitting the stages across additional model calls costs

more and lowers test success. It does not establish a general claim about multi-agent systems. It also does not isolate a universal benefit of role names: the label effects remain small, sensitive to the surrounding instructions and unconfirmed in the corrected quality tests.

8 Initial factorial study with GPT-5.4 mini

8.1 Allocation, observed outcomes and missingness

All 1,000 assignments and 1,800 planned turns were submitted. The original attempt completed 457 of 460 submitted cells; the separately frozen continuation completed 539 of 540 untouched cells. Four cells remain incomplete: D on /1028 and SN on /388 after transport errors, SN on /249 after a deadline in the original attempt, and SN on /1028 after the continuation deadline. The final interrupted stream includes a completed response and usage, but its terminal status violates the 180-second acceptance criterion. It is therefore not used as an observed program. Event arrival times are not independently timestamped, and no timing sequence beyond the recorded terminal result is inferred.

The native evaluator checked all 996 observed programs. After the common reference gate, 963 observations remain evaluable: 453 passes, 507 failures and three native timeouts. The other 33 observed programs belong to reference-ineligible tasks. Two final-format extraction failures, D /494 and MR /100, remain empty observed programs with their original responses retained. Generation failures, native timeouts, format errors and unavailable controls are distinct in the published records and in the complete task appendix.

Table 6 reports observed rates and extreme all-assignment ranges. Across the five primary conditions, observed test-success rates range from 44.56% to 48.70%, giving substantial room for both improvement and deterioration. The lower SN denominator reflects two missing generations on otherwise eligible tasks; both /1028 missing generations coincide with an unavailable reference control. The INoT* row belongs to a separate exploratory batch, discussed below.

Table 6 Main sample: 200 assignments per condition. Quality uses eligible observed programs; ranges retain all 200 assignments and are not confidence intervals. INoT* is a separate exploratory algorithm replication.

Arm	Generated	Passes	Rate (%)	Range (%)	Format errors
D	199	94/193	48.70	47.00–50.50	1
SN	197	93/191	48.69	46.50–51.00	0
SR	200	92/193	47.67	46.00–49.50	0
MN	200	86/193	44.56	43.00–46.50	0
MR	200	88/193	45.60	44.00–47.50	1
INoT*	199	96/192	50.00	48.00–52.00	0

8.2 Four predeclared quality contrasts

Table 7 and Fig. 4 show paired differences. Single-call roles produce six wins and eight losses against neutral stages on 191 eligible pairs. Three-call roles produce ten wins and eight losses on 193 pairs. Neither comparison provides evidence of improved quality after correction. The two topology contrasts also fail to reject: the smallest adjusted p is 0.3134 for MN versus SN. The descriptive difference between the three-call and one-call role contrasts is +2.09 percentage points, with interval $[-4.19, +8.38]$, on the common 191-task subset.

These findings do not establish equivalent quality. In particular, the interval for SR minus SN still includes a loss of 4.71 points. The resource reduction therefore does not establish savings with preserved quality. Failure to reject the other contrasts similarly leaves improvements and losses compatible with the observed uncertainty.

Table 7 Four predeclared quality comparisons under amended execution. Differences and descriptive 95% task-bootstrap intervals are percentage points. W/L are discordant pairs in favour of the first/second condition. Holm adjusts the exact two-sided McNemar tests across all four comparisons.

Contrast	Pairs	Difference [interval]	W/L	p	p_{Holm}
SR - SN	191	-1.05 $[-4.71, +2.62]$	6/8	0.7905	1.0000
MR - MN	193	+1.04 $[-3.11, +5.70]$	10/8	0.8145	1.0000
MN - SN	191	-4.71 $[-9.42, +0.00]$	6/15	0.0784	0.3134
MR - SR	193	-2.07 $[-6.22, +2.07]$	7/11	0.4807	1.0000

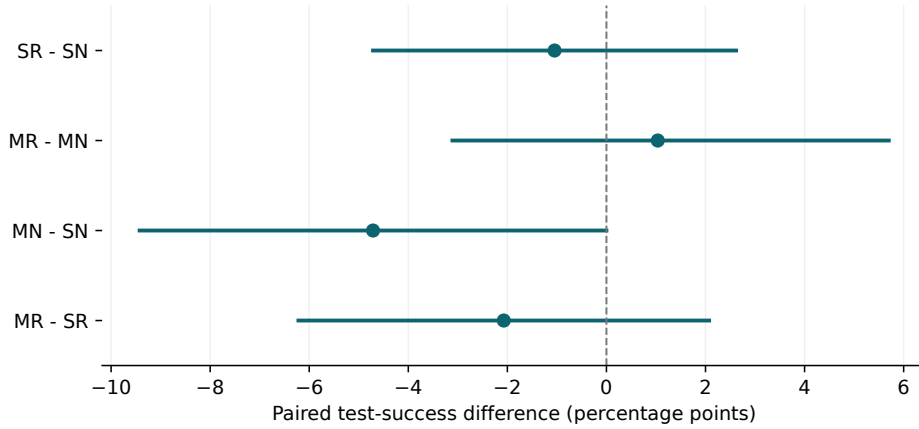


Figure 4 Paired quality differences in percentage points, with descriptive 95% task-bootstrap intervals. The reference line is zero; Holm-adjusted tests are reported separately in Table 7.

8.3 Measured resources and price sensitivity

Completed candidates use 8,569,456 tokens and have total valuation USD 14.1467676. The submitted-turn ledger additionally retains 15,729 known tokens from the interrupted continuation call. Across all submitted calls, known usage is therefore 8,585,185 tokens: 5,776,714 input, including 4,097,536 cached, and 2,808,471 output, including 2,076,742 reasoning. The 1,797 calls with counters are valued at USD 14.2048182, or USD 16.970655 without cache discounts. Three interrupted calls have unknown usage; the known total is not a complete account of consumption. None of these valuations is a subscription charge.

Table 8 reports per-condition means over completed candidates. The paired comparisons in Table 9 use a common subset for each resource contrast. SR versus SN uses 197 pairs: 8.76% fewer tokens and 18.45% lower valuation, with a mean USD difference of -0.0019761 [-0.0027617, -0.0012296]. The no-cache valuation remains 16.21% lower. Thus the observed label-associated resource difference does not disappear when the cache discount is removed, although its magnitude changes. These resource intervals are descriptive and outside the four quality tests.

Within three calls, the MR/MN valuation ratio is 0.985 and its mean-difference interval crosses zero. The single-call resource finding therefore does not generalize clearly across topologies. Conversely, MN/SN and MR/SR use 2.84 and 3.06 times as many tokens and cost 2.09 and 2.50 times as much; no-cache valuation ratios are 2.19 and 2.56. Full repeated inputs and intermediate responses form part of this measured expense. Figure 5 separates cached input, uncached input and output without double-counting reasoning.

Table 8 Resource means over completed candidates, including reference-ineligible tasks. Costs are API-equivalent USD; no-cache valuation removes the input cache discount. Interrupted-attempt usage is additionally reported in the submitted-turn ledger.

Arm	Candidates	Tokens	USD	No cache, USD
D	199	4 160	0.00655	0.00814
SN	197	5 122	0.01071	0.01230
SR	200	4 703	0.00888	0.01044
MN	200	14 578	0.02257	0.02714
MR	200	14 382	0.02222	0.02676
INoT*	199	3 858	0.00532	0.00670

Sensitivity to missing outcomes and execution batches

A post hoc diagnostic uses retained records only. Batch membership was recovered from cell IDs in the continuation manifest: 460 assigned cells are original and 540 are continuation; 996 candidate records are retained and four submitted-incomplete

Table 9 Descriptive paired resource contrasts. Ratios divide paired means; uncertainty is a 95% task-bootstrap interval for the paired mean difference in API-equivalent USD, not for the ratio.

Contrast	Pairs	Token ratio	USD ratio	USD difference [interval]
SR - SN	197	0.912	0.816	-0.0020 [-0.0028, -0.0012]
MR - MN	200	0.987	0.985	-0.0003 [-0.0016, +0.0008]
MN - SN	197	2.839	2.088	+0.0117 [+0.0104, +0.0130]
MR - SR	200	3.058	2.504	+0.0133 [+0.0121, +0.0147]

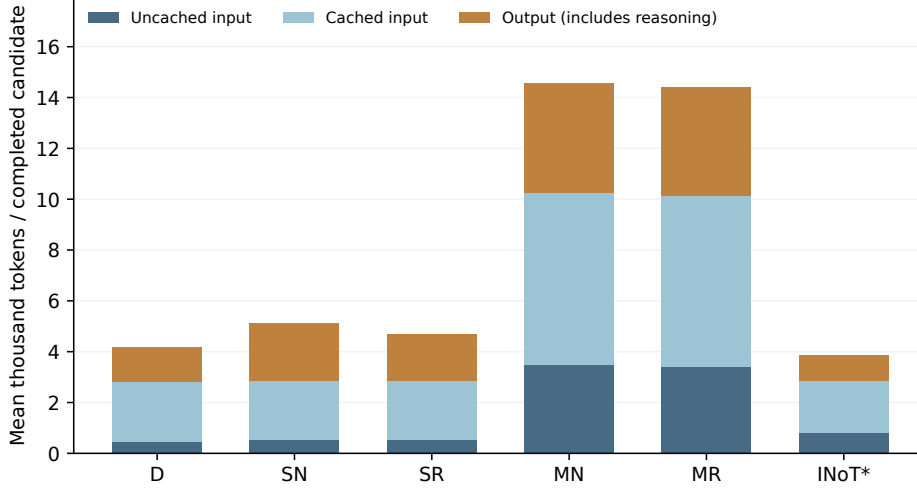


Figure 5 Mean token components per completed candidate. Cached tokens are separated from the remaining input; output includes reasoning. Denominators include reference-ineligible tasks. INoT* is a separately dispatched adaptation.

cells are unavailable. First-batch-only estimates are therefore not treated as an unbiased counterfactual for a complete original execution. The bounds table reports finite-sample identification limits for unknown binary endpoints, while the batch table reports descriptive within-batch contrasts on complete task pairs. Task bootstrap intervals quantify conditional task-sampling uncertainty in those observed subsets; they do not quantify provider or time variation, batch assignment, or missingness mechanisms. The cross-batch complete-pair counts are 2, 1, 0, and 1 for SR-SN, MR-MN, MN-SN, and MR-SR, respectively; singleton bounds remain sampling-uncertain despite having zero identification width. These counts do not identify provider drift or a causal batch effect. This supplement does not alter the four predeclared McNemar tests or their Holm correction.

Table 10 Missingness identification bounds (percentage points).

Contrast	193 control-eligible tasks	All 200 assigned tasks
SR–SN	$[-1.55, -0.52]$	$[-5.00, +3.00]$
MR–MN	$[+1.04, +1.04]$	$[-2.50, +4.50]$
MN–SN	$[-4.66, -3.63]$	$[-8.00, +0.00]$
MR–SR	$[-2.07, -2.07]$	$[-5.50, +1.50]$

Unknown endpoints range independently over 0 and 1; bounds are identification ranges, not confidence intervals.

Table 11 Within-batch quality sensitivity (percentage points).

Batch	Contrast	N	Difference	Descriptive task-bootstrap 95% interval
Original	SR–SN	87	-2.30	$[-8.05, +3.45]$
Original	MR–MN	89	+3.37	$[-3.37, +11.24]$
Original	MN–SN	87	-2.30	$[-9.20, +4.60]$
Original	MR–SR	90	+4.44	$[-2.22, +11.11]$
Continuation	SR–SN	102	+0.00	$[-4.93, +4.90]$
Continuation	MR–MN	103	-0.97	$[-5.83, +3.88]$
Continuation	MN–SN	104	-6.73	$[-13.46, +0.00]$
Continuation	MR–SR	102	-7.84	$[-13.73, -2.94]$

Intervals quantify conditional task-sampling uncertainty for the observed complete-pair subset; they exclude provider/time variation and missingness mechanisms. Native timeouts are observed failures under the frozen rules. The four primary McNemar/Holm tests are unchanged.

9 Exploratory comparators and development replication

Direct solving has the lowest observed token and valuation means among the five main conditions. SN minus D has a paired quality difference of zero $[-4.19, +4.19]$ on 191 tasks, but a valuation ratio of 1.632 on 197 resource pairs. MR minus D has a quality difference of -3.11 $[-7.77, +1.04]$ and costs 3.359 times as much on 199 resource pairs. These descriptive comparisons support keeping a direct baseline; they do not establish statistical dominance.

The separately dispatched INoT* adaptation yields 199 complete programs from 200 assignments. One assignment reaches the original deadline; none fails final-format extraction. On 192 eligible outcomes, 96 pass (50%); all-assignment bounds are 48–52%. Known usage is 767,679 tokens and USD 1.05886035 valuation (USD 1.33326675 without cache discounts), with one call lacking usage. Relative to D, its paired quality difference is +1.04 $[-3.65, +5.73]$ and its valuation ratio is 0.805; relative to SR the corresponding values are +2.08 $[-2.08, +6.25]$ and 0.600. Table 12 retains different quality/resource pair counts. Because this algorithm adaptation ran in a separate batch, timing, cache and provider changes remain potential confounders. It is not an author-code replication or part of the four-test family. Figure 6 places the observed means on a common scale without declaring a statistically established frontier.

Table 12 Exploratory comparisons outside the four-test family. Intervals are descriptive 95% task bootstraps. Quality/resource columns give their respective complete-pair counts. INoT* was dispatched in a separate batch.

Contrast	Pairs Q/R	Quality difference [interval]	Tokens, ratio	USD, ratio
SN - D	191/197	+0.00 [−4.19, +4.19]	1.230	1.632
MR - D	193/199	−3.11 [−7.77, +1.04]	3.444	3.359
INoT* - D	192/198	+1.04 [−3.65, +5.73]	0.925	0.805
INoT* - SR	192/199	+2.08 [−2.08, +6.25]	0.821	0.600

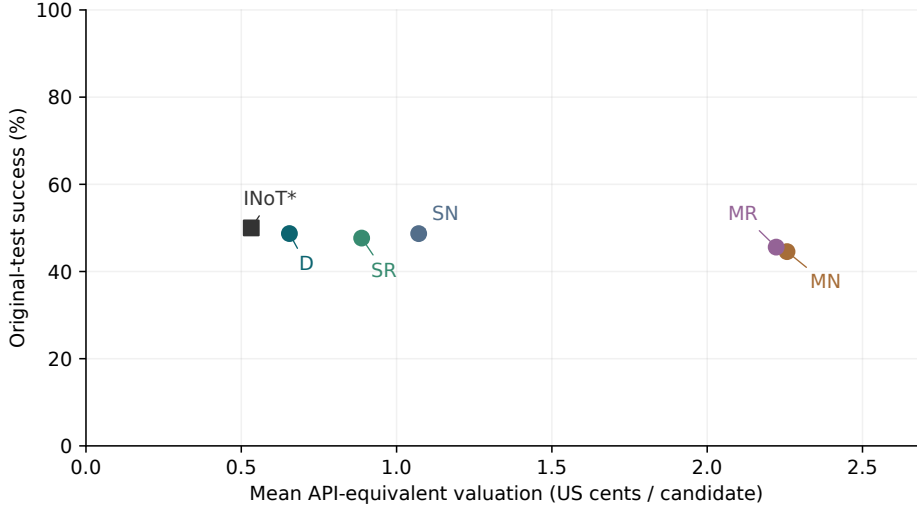


Figure 6 Observed test-success rates and mean API-equivalent valuation. The full 0–100% quality axis avoids exaggerating small rate differences. These points have different eligible denominators; paired uncertainty is provided in Tables 7 and 12.

10 Separate development calibration

The development stage precedes the main sample and uses 40 disjoint randomly selected tasks, the same five conditions, and two fresh labelled repeats (2 and 3). All 400 assigned programs and 720 CLI turns complete and pass the trace audit. All 400 native records match the exported programs exactly. Reference task /1005 fails in the development environment, leaving 390 evaluable attempts over 39 tasks; its ten generation costs remain included. No final-format extraction fails. Each attempt is scored separately, with no best-of-two selection. For paired development contrasts, the two repeats are averaged within task before resampling.

The observed rates range from 43.59% to 53.85%. D passes 42/78 attempts, SR and MN each 39/78, MR 36/78, and SN 34/78. D also has the smallest resource mean. Complete generation uses 3,703,821 tokens, valued at USD 6.6113451, or USD 7.7687595 without cache discounts. The denominator remains 40 task clusters for resources and 39 for quality contrasts.

Table 13 Development results: 80 assigned candidates per condition, two repeats on 40 tasks. Quality uses 78 evaluable attempts; resource means use all 80. Missingness ranges use the full assignment denominator and are not confidence intervals.

Arm	Passes	Rate (%)	Range (%)	Tokens	USD	No cache
D	42/78	53.85	52.50–55.00	4 500	0.00776	0.00939
SN	34/78	43.59	42.50–45.00	5 598	0.01251	0.01417
SR	39/78	50.00	48.75–51.25	5 189	0.01070	0.01234
MN	39/78	50.00	48.75–51.25	15 636	0.02624	0.03090
MR	36/78	46.15	45.00–47.50	15 375	0.02543	0.03030

Table 14 Paired development contrasts. Quality differences and descriptive 95% task-bootstrap intervals are percentage points over 39 complete task clusters; resource ratios use 40. The last two comparisons are exploratory. No confirmatory p-value or non-inferiority test is assigned.

Contrast	Quality difference [interval]	Token ratio	USD ratio
SR - SN	+6.41 [+1.28, +11.54]	0.927	0.855
MR - MN	−3.85 [−11.54, +2.56]	0.983	0.969
MN - SN	+6.41 [−2.56, +15.38]	2.793	2.097
MR - SR	−3.85 [−8.97, +1.28]	2.963	2.377
SN - D	−10.26 [−19.23, −2.56]	1.244	1.613
MR - D	−7.69 [−17.95, +1.28]	3.417	3.277

The role-minus-neutral development differences are +6.41 percentage points in one call and −3.85 in three calls. The difference of differences is −10.26 points, with a descriptive task-bootstrap interval of [−19.23, −1.28]. Three-call token means are 2.79 and 2.96 times the corresponding single-call means; valuation ratios are 2.10 and 2.38. These observations motivated the separate reserved-task allocation. They are not additional observations in its four-test family and cannot replace a main-sample result. Complete two-repeat outcomes are provided in the appendix.

11 Illustrative discordant cases

For each label contrast, we selected the lowest numeric task ID with an observed, eligibility-gated discordance in each direction. Thus, the four examples contain one role-labelled–neutral discordance favouring each condition for each contrast. They are descriptive examples outside the four primary tests. They do not estimate discordance frequencies or identify a hidden reasoning mechanism.

For `BigCodeBench/428`, the role-labelled single-call program passed while the neutral program failed. Both programs performed the requested outer merge and scaling of numeric features from `df1`. The role-labelled program scaled a copy of `df1` before one merge and preserved the left-dataframe columns first. The neutral program merged

first, then dropped and reinserted the scaled columns in a second merge, producing the order `id`, `feature4`, `feature5`, `feature1`, `feature2`, `feature3`. The evaluator’s `test_case_1` required `id`, `feature1`, `feature2`, `feature3`, `feature4`, `feature5` and failed with that list mismatch. The task prose specifies the operations but does not explicitly specify column order; the test supplies a stricter output contract.

For `BigCodeBench/71`, the neutral single-call program passed while the role-labelled program failed. The role-labelled implementation used sample standard deviation, `Series.std(ddof=1)`, while the neutral implementation and canonical solution used population standard deviation, `np.std` with `ddof=0`. The role-labelled program consequently failed `test_case_2`, `test_case_3`, `test_case_4`, and `test_case_5`; the recorded discrepancies were respectively 11.900380145988935 versus 11.017611134090766, 8.568005268772557 versus 8.01463505095522, NaN versus 0.0, and 47.95684491664328 versus 46.07544623291379. “Standard deviation” is underspecified in the prose, while the executable contract fixes the population convention.

For `BigCodeBench/167`, the role-labelled three-call program passed while the neutral program failed. The neutral implementation selected `max(2,min(5,num_types))` stacked segments. It therefore produced 50 patches for `num_types=10` and 2 for `num_types=1`, whereas the role-labelled implementation used exactly `num_types` segments and produced 100 and 1. The native failures were `test_case_3`, `50  $\neq$  100`, and `test_case_4`, `2  $\neq$  1`. The prose describes categories and segmented values but leaves the segment count open; the tests require it to equal `num_types`.

For `BigCodeBench/31`, the neutral three-call program passed while the role-labelled program failed. The neutral implementation preserved the first-occurrence order of the `Counter`; the role-labelled implementation sorted by descending frequency and then lexicographically. On the test text containing `$is` three times, `$second` once, and `$sentence` twice, the evaluator expected values 3, 1, 2 in first-occurrence category order. Sorting moved `$sentence` ahead of `$second`, and `test_case_2` reported “Expected value ‘1.0’, but got ‘2’.” The prose requires a frequency plot without specifying category order, so this is an executable-order mismatch rather than evidence of a causal role effect.

All descriptions use the recorded prompts, programs, and native assertions. They describe observed code behavior only; they do not infer private reasoning, latent agents, or representative population frequencies.

12 Independent boundary-robustness experiment

The main SR–SN contrast operationalizes role organization through label substitution. An additional prospective experiment tests whether this comparison persists under a second, explicitly controlled instruction layout. It uses one fresh response in every condition, crossing role labels (planner, implementer, reviewer versus stage 1, stage 2, stage 3) with bracketed headings on separate lines versus colon-labelled sentences in one paragraph. Within each layout, only the three labels change; across layouts, the operation sentences and their order remain identical. The common instruction requests internal analysis, implementation and review and prohibits checkpoint commentary in

the answer. Every candidate must be one complete fenced Python program. Neither the response topology nor compression varies.

This intervention isolates the specified role-label wording conditional on each layout. It cannot establish that the model executed separate internal roles: checkpoint compliance and private computation remain unobserved. The layout comparison concerns visible instruction formatting, not independent agents or separated information access. A persistent label effect across both layouts would strengthen robustness for this implementation; a changed direction or a wide interval would constrain that interpretation. The additional data are not pooled into the original four-test family.

The sample comprises the next 80 task IDs in the previously randomized order of the reserved pool, excluding all 200 main tasks, 40 development tasks and eight exposed dataset-card examples. Selection is mechanical and precedes generation. Each task receives all four conditions, giving 320 assignments and at most 80 paired task units. A frozen task-block order interleaves the four conditions with four concurrent calls. GPT-5.4 mini, medium reasoning, Codex CLI 0.153.4, fresh ephemeral subscription calls, full supplied context and disabled tools are unchanged. Each call has a 600-second deadline and the same 65,536-byte input guard; no prompt is shortened and no candidate is retried. These deadlines differ from the main series and the additional estimates are reported separately.

The native BigCodeBench source and tests are unchanged. Before generation, missing dependencies and older library contracts are addressed in separately recorded container builds and full-allocation control runs. The final image passes all 80 reference programs; the incorrect control fails on 79 tasks but passes on BigCodeBench/59. That task remains assigned and evaluated in every condition, but its quality is control-unavailable. The final gate therefore admits at most 79 quality pairs. All failed control attempts, package versions and image identities are retained; no reference program, test or task ID is replaced. The generation processes never receive these evaluator artifacts.

The prespecified primary family comprises two exact two-sided McNemar tests: role versus neutral within headings and within prose, with Holm correction at familywise $\alpha = 0.05$. Heading-versus-prose contrasts and the difference of these two role contrasts are descriptive. Task bootstrap intervals use 10,000 resamples and analysis seed 20260908. Resource contrasts report paired token differences, API-equivalent valuations and ratios of paired means; no equivalence, non-inferiority or joint quality-cost decision is defined. A pre-generation exact power calculation at the conservative first Holm threshold 0.025 records sensitivity to discordance and effect size. The sample does not provide high power to rule out small effects; this limitation is retained regardless of the observed outcome. The complete execution and analysis plan was fixed before dispatch.

12.1 Complete evaluation and label effects

All 320 assignments yielded complete programs and native outcomes, with no retry, missing result or evaluation timeout. Of the 316 control-eligible outcomes, 173 pass and 143 fail. The four outcomes on /59 remain visible in the complete task appendix

Table 15 Independent 80-task extension. RB/NB: role/neutral labels with headings; RP/NP: the same labels in prose. V is mean standard API-equivalent valuation per observed candidate, not an invoice.

Arm	Generated	Passed/eligible	Pass (%)	Tokens	V (USD)
RB	80/80	44/79	55.70	4415	0.007244
NB	80/80	44/79	55.70	4431	0.007320
RP	80/80	42/79	53.16	4331	0.006881
NP	80/80	43/79	54.43	4331	0.006884

Table 16 The two prespecified role contrasts. Differences and descriptive task-bootstrap 95% intervals are in percentage points. W/L counts discordant pairs; Holm correction covers both exact two-sided McNemar tests.

Contrast	Pairs	Difference [interval]	W/L	p	p_{Holm}
RB - NB	79	+0.00 [-6.33, +6.33]	3/3	1.0000	1.0000
RP - NP	79	-1.27 [-10.13, +6.33]	5/6	1.0000	1.0000

(three passes and one failure), but do not enter quality inference. Table 15 gives the four condition totals.

With headings, roles and neutral stages both solve 44 of 79 eligible tasks (55.70%); there are three paired wins and three losses. In prose, roles solve 42 tasks (53.16%) and neutral stages solve 43 (54.43%), with five wins and six losses. The role differences are respectively 0.00 percentage points, descriptive 95% interval [-6.33, +6.33], and -1.27 [-10.13, +6.33]. Both exact two-sided McNemar tests have raw and Holm-adjusted $p = 1$ (Table 16). These wide intervals neither establish equivalence nor exclude a meaningful quality effect.

The descriptive heading-minus-prose differences are +2.53 points [-5.06, +10.13] with roles and +1.27 [-6.33, +8.86] with neutral labels. Their interaction is +1.27 [-10.13, +12.66]. Thus neither the role comparisons nor the layout estimates establish a quality advantage. This is a test of observable instructions, not a measurement of whether independent internal roles actually occurred.

12.2 Resources and relation to the primary experiment

All 320 calls have usage counters: 932,812 input tokens, including 797,696 cached, and 467,794 output tokens, including 373,494 reasoning. Total usage is 1,400,606 tokens, with standard API-equivalent valuation USD 2.2662372 and no-cache sensitivity USD 2.804682. These are valuations of subscription calls, not API payments.

For roles versus neutral stages under headings, the token ratio is 0.9964 and the valuation ratio is 0.9897. The paired mean valuation difference is USD -0.0000756, with descriptive interval [-0.0011090, +0.0008682]. In prose, the corresponding ratios are 0.9998 and 0.9996, with valuation difference USD -0.0000029 [-0.0011391, +0.0011926].

Without cache discounts the valuation ratios are 0.9940 and 1.0022; neither resource-difference interval excludes zero. Figure 7 displays all four contrasts.

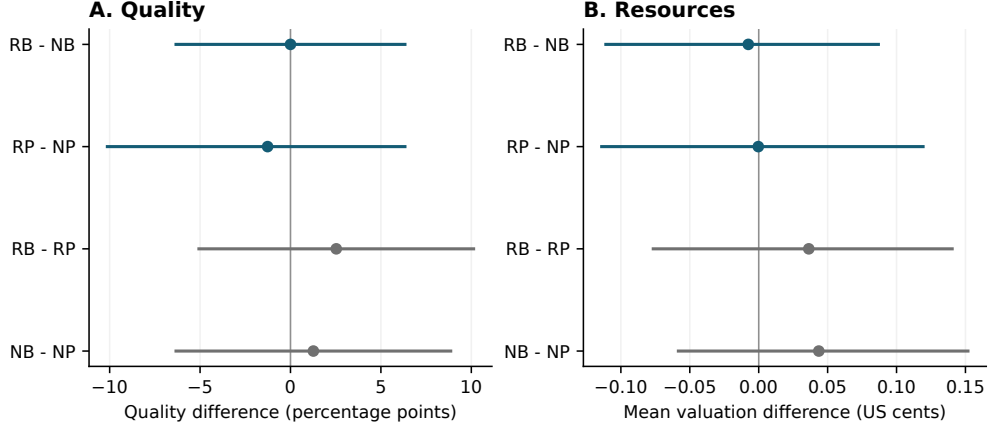


Figure 7 Independent experiment: paired quality differences (percentage points) and API-equivalent valuation differences (US cents per candidate), with descriptive 95% task-bootstrap intervals. The first two contrasts form the primary quality-test family; the layout contrasts are descriptive. RB/NB denote role/neutral headings, RP/NP role/neutral prose.

The additional wording-and-layout experiment therefore does not convincingly reproduce the primary series’ 18.45% lower valuation for the single-call role condition. This comparison across batches is descriptive: task sample, precise instructions, execution time, environment and deadline differ, and no formal between-study heterogeneity test is claimed. The new estimates limit generalization of the earlier resource finding to role names alone. They do not invalidate the recorded primary observations, prove zero effect, or test the three-call topology. The two prospective analyses remain separate.

13 Complete repository-patch demonstration

A separate one-task SWE-bench Lite development demonstration uses `pvlb__pvlb-python-1606`. Deterministic BM25 retrieval from the benchmark-designated initial software version selects whole eligible Python files within 24,000 source bytes; nine files totaling 23,863 bytes enter the identical packet for all arms. Test files, hints and reference patches are excluded from retrieval. Files too large to fit are listed, not truncated. This is a selected packet, not the full repository; retrieval omissions remain a limitation.

The unchanged pinned SWE-bench harness initially fails even with the gold patch because its official task image contains an incompatible NumPy version. A separate image changing NumPy to 1.26.4 passes all 11 gold tests; an applicable comment-only negative passes ten tests and fails the regression test. Both controls are repeated successfully before evaluation of the additional candidates. Initial failures and image identities are retained. The controller runs offline, while the unchanged upstream child

Table 17 All ten SWE development assignments on one issue. A denotes native patch-application rejection; T denotes an applied patch that fails the target regression while passing the other ten tests. Every entry is an observed failure, not missing data. Repeat labels are not controlled model seeds.

Condition	Repeat 1	Repeat 2
D	A	T (original)
SN	A	A (original)
SR	A	A
MN	T	A
MR	A	A (recovery)

task container retains its default Docker network; no credentials or personal directories are mounted.

The original ten-assignment generation stopped after a 180-second CLI deadline, leaving two complete candidates, seven unsubmitted assignments and one interrupted MR assignment. Before additional generation, a completion protocol specifies one attempt for each of the seven untouched assignments. It also restarts the interrupted MR assignment from its first stage as a separately labelled recovery. Each additional turn receives 600 seconds. No candidate is repaired, selected or regenerated using test feedback. The original interrupted attempt remains in the first-attempt history and resource ledger. The completed table therefore combines two original candidates and eight extension candidates generated under different operational deadlines; it is not a homogeneous first-attempt estimate.

For all ten assignments, the evaluated patch is checked against the patch extracted from the model response (Table 17). None resolves the issue: eight patches fail application and two apply but fail the target regression. No evaluated candidate has an identified infrastructure failure. This supports end-to-end evaluation feasibility and exposes a patch-generation failure mode; it does not support repository-repair effectiveness or an advantage of role instructions. Ten attempts on one issue are one task cluster, not ten independent benchmark instances.

Patch-serialization errors limit the interpretation of this negative result. Four additional responses use the tool-specific ***** Begin Patch** syntax, and all ten patches contain a bare @@ hunk marker. These observations do not establish the functional quality of every intended edit. Because the frozen fence extractor strips the final new-line, a separately specified post-observation sensitivity appends exactly one final LF to all ten predictions and reevaluates them under fresh run IDs. All ten classifications remain unchanged. This diagnostic changes no program, hunk header or file context and does not replace the original outcomes.

The extension completes all 16 planned turns, consuming 314,793 known tokens valued at USD 0.758976. The original run submitted four turns, three with known usage and one unknown. Across both phases, 19 of 20 submitted turns expose usage: 371,589 known tokens and USD 0.8969655 in standard API-equivalent valuation, including the interrupted attempt’s known first stage. Unknown usage is not zero, and this valuation is not a subscription invoice.

The absence of an instance report is not itself an outcome. An observed response yielding an empty patch is a candidate failure. A nonempty patch is a failure when the

exact exported bytes, native run summary and upstream patch-application rejection evidence agree. Other missing reports, explicit infrastructure failures and inconsistent response-export joins remain unknown. The ten classifications are checked against the original responses, extracted patches, control results and evaluator reports. No SWE-bench Lite score or cross-task generalization claim follows from this demonstration.

14 Discussion

The larger factorial study separates two findings. Under this protocol, moving the prescribed stages across three calls has a substantial resource effect and a quality cost relative to keeping them in one response: the three-call variants consume more and pass fewer benchmark tests than the corresponding single-call variants. The smaller role-label differences do not support corrected quality rejections. A shared response can therefore be a useful implementation choice without its named roles providing an independently demonstrated quality benefit.

The initial mini study and the separate label-by-layout experiment retain their own samples, execution settings and statistical families. The initial single-call role-associated valuation reduction is larger than the Luna result; the minimal label-by-layout study does not convincingly reproduce that reduction. These comparisons limit claims about robustness. They are not a matched experiment on model identity or a formal test that an effect changed across studies. Repeated calls within a task improve its success-rate estimate but do not add new task units or resolve public-benchmark training overlap.

The theoretical resource balance remains conditional. Repeated context transmission is one measurable expense, while output length, provider reasoning and caching also respond to the instructions. The experiment therefore measures the implemented configurations rather than an optimal allocation under a common hard inference budget. It does not show that multiple calls cannot help in an interactive repository setting.

An algorithmic instruction such as INoT* and a minimal role-label substitution are different interventions. The former changes the reasoning procedure; SCC additionally changes interaction and generated-test feedback. Their comparisons must retain those scopes. Likewise, benchmark test success measures the specified executable contract. The development cases show why prompt-test agreement deserves separate scrutiny, and their outcomes are retained rather than excluded after inspection.

14.1 What the evidence means for the architecture

The architectural result concerns where the prescribed work is placed. SR retains the planning–implementation–review instruction in a single call and uses fewer observed resources than its three-call counterpart MR. The SR–SN contrast answers a different question: it changes the role labels while holding the operations and call count fixed. The lower SR resource mean does not establish a quality benefit from role segregation. Moreover, D uses fewer resources than SR within the main factorial batch. The

present study evaluates these choices as components of the Hybrid-INoT core; it supports studying a compact internal configuration, but does not establish that its role structure is necessary.

External verification defines the reported outcome: it distinguishes a generated program from one that passes the executable contract. Every condition receives the same evaluator, so the evaluator cannot explain an advantage unique to Hybrid-INoT. An offline test result also does not show that a deployed agent can autonomously repair a failure. The separate INoT* comparison extends the question to a different internal-dialogue instruction; it does not change the definition of the SR core.

14.2 Design extensions and their evidence requirements

Table 18 Architectural questions and the scope of the current evidence. These organize the discussion; they are not additional confirmatory tests.

Question	Current contribution	Evidence still required
Shared-context efficiency with preserved quality	Conditional token balance; component quality/resource comparisons without compression	A controlled context-length intervention and externally justified quality-preservation criterion
Economic value of preliminary screening	A prospective break-even condition	Integrated screening/rerun policy with final verified outcomes and all costs
Boundary under parallel tools	Explicitly outside fixed text-only generation	Matched serial/parallel tool workflows on independent subtasks; latency and verified quality

For example, let a small preliminary checker cost c_s , let an expensive rerun have constant cost c_r , and let rerun probabilities under two policies be ρ_0 and ρ_1 . Under exactly one preliminary check and at most one rerun, the change in expected inference cost is

$$\Delta C = c_s + (\rho_1 - \rho_0)c_r. \quad (6)$$

The augmented policy costs less exactly when $c_s < (\rho_0 - \rho_1)c_r$. This identity states the economic condition but does not estimate the policy effect: a checker can reduce reruns simply by accepting wrong candidates. A useful policy also needs verified quality and a prospectively justified acceptance rule. No small-model screening or rerun policy is executed in the primary study.

Independent tools may likewise benefit from external parallelism even when a single role call uses fewer tokens. The present serial stage sequence does not test latency. Compression, adaptive mode selection, and verification-driven correction would each alter the intervention and require a separate comparison. They are possible extensions of Hybrid-INoT, not validated components of the reported resource difference.

14.3 Author-code SCC implementation feasibility

A separate development pilot executes the original Self-collaboration role classes and session controller from the 2024 author implementation, using a disclosed Codex subscription transport adaptation with GPT-5.6 Luna and medium reasoning. Three previously exposed development tasks (/325, /322 and /1036), selected by their existing order, produce three complete candidates in 11 model calls. All three candidates undergo native BigCodeBench evaluation: /325 passes; /322 and /1036 fail. All three reference programs pass and all three incorrect controls fail in the same pinned environment. Known usage totals 48,023 input-plus-output tokens, with no missing call counters, and USD 0.0138646 in API-equivalent valuation. This is neither subscription billing nor an estimate of full evaluation cost.

The Luna valuation uses its own registered base rates, USD 0.20/0.02/1.20 per million input/cached-input/output tokens [13]:

$$V_{\text{Luna}} = \frac{0.20(I - C) + 0.02C + 1.20O}{10^6}.$$

Here C is included in I , and reasoning tokens in O . These coefficients differ from the earlier mini series. The formula describes base-rate valuation; long-prompt and cache-write surcharges require separate scope checks.

The 2023 paper describes a tester that simulates testing and returns a natural-language report [1]. The pinned 2024 author code instead executes model-generated examples and treats execution without an exception as its internal stopping signal; printed values are not automatically compared with expected answers. Our reproduction target is this code version [2]. The controller logic, prompts and extraction are retained, with a cap of two coder iterations rather than the source constructor’s default of four: one initial implementation and at most one repair, requiring at most four model calls including the analyst and tester. This cap limits the comparison to that configuration. Generated code executes in an isolated container. The adapter serializes role messages through the CLI and does not reproduce the original API sampling controls; it supplies the full task as text and expects self-contained returned code. The retained responses were checked against the session transitions and all three final programs. This initial pilot has one repetition and no concurrent comparator arms.

A subsequent development gate on 11 September 2026 executes fresh SR, SN and SCC assignments on the same three exposed tasks using the common Luna transport. All nine programs receive native evaluation: SR passes 0/3 tasks, SN 1/3 and SCC 1/3. All 18 model calls have retained responses and usage counters; their known API-equivalent valuation totals USD 0.02301176. The control records show reference passes and incorrect-program failures in the unchanged native environment; the final SCC programs match the controller transitions recorded for this pilot. This gate checks the executable comparison procedure, including failures; three exposed tasks and one repetition do not support an inferential method comparison. Both pilots are excluded from the 1,000-task allocation.

14.4 Task similarity in the expanded allocation

Different task identifiers do not necessarily imply different source material. An additional similarity audit was therefore specified for the Luna extension in Appendix B.5. It covers all 1,140 BigCodeBench source tasks and the fixed allocation of 1,000 tasks. Only source instructions and reference programs are used; generated answers and evaluation outcomes are excluded.

Tasks are linked using three criteria:

1. **Exact matches.** Instructions are compared after whitespace normalization. Reference programs are compared by their abstract syntax trees: docstrings are removed, while names, literal values and code structure are preserved. All selected instructions are distinct after whitespace normalization. However, reference programs for tasks 130 and 131 have the same syntax tree.
2. **Instruction similarity.** The starter-code paragraph is removed and letter case is normalized. Sets of five consecutive words are then compared. Their Jaccard coefficient is the fraction of shared elements in their union; values closer to one indicate greater similarity.
3. **Code similarity.** Each program must contain at least 20 identifier occurrences. The set Jaccard similarity must be at least 0.80 and the multiset Jaccard similarity at least 0.70; both measures use non-keyword identifiers and string or numeric literals. Multisets account for repetition. Keyword tokens are excluded; the spelling of all other tokens is preserved.

This is an adaptation of token-based duplicate screening [20]. It detects specified forms of similarity rather than establishing semantic equivalence.

Links are first constructed across all 1,140 source tasks. Tasks connected directly or through other tasks form a group. Each group is then intersected with the assigned sample. The combined rule uses exact matches, code similarity and instruction similarity with a threshold of 0.70. It yields 992 groups: seven are non-singleton groups containing 15 tasks in total, and the remaining groups are singletons. The largest group contains three tasks. Table 19 lists all seven groups. Instruction-only thresholds of 0.50, 0.70 and 0.90 yield 996, 999 and 1,000 groups, respectively. These counts describe screening partitions; they do not establish semantic independence.

There is also overlap across studies. Task 1120 has the same instruction and reference syntax tree as task 1121 in the separate 80-task experiment. Task 826 has the same reference syntax tree as development task 814. Disjoint task identifiers therefore do not guarantee disjoint source material.

No tasks are excluded from analysis. The supplementary interval sensitivity analysis first averages three repeated differences within each task, then resamples whole groups with replacement 10,000 times using seed 20260911. Each draw divides the sum of sampled task differences by the sampled task count. A larger group therefore does not receive the same weight as a single task. The resulting 95% percentile intervals depend on the grouping rule and are reported separately from the primary tests. For the completed Luna extension, this analysis uses the observed three-repeat task

Table 19 Task groups under the combined similarity rule. The second column contains BigCodeBench task numbers. For example, 1097 denotes BigCodeBench/1097.

Group	BigCodeBench task numbers	Task count
1	1097, 1099	2
2	130, 131	2
3	349, 351, 353	3
4	369, 622	2
5	401, 413	2
6	423, 426	2
7	673, 675	2

differences and fixed source partitions. SCC source partitions are frozen, but its generation and native outcome inspection remain ongoing; no SCC outcome interval is reported. Unequal group sizes and unrecognized dependence limit interpretation [21].

15 Validity and limits of generalization

The initial and expanded factorial studies use two requested model aliases, each at one reasoning setting and one client version. Their samples and replication counts differ. This provides more scope than a single model run but does not identify model effects or generalize across model families. The development repeats are a separate feasibility study. Public task availability leaves training contamination unresolved; related tasks can also violate an independent-task approximation. A different model, language, task distribution or service revision could reverse an observed contrast. The recorded client version and execution dates do not fix the provider’s model weights.

The deadline and administrative continuation create a second boundary. Failed generations remain missing, and the amendment is disclosed. Inference from available pairs concerns the observed eligible subset. Assignment-denominator ranges show extreme missing-outcome possibilities but do not correct selection bias. A time-out could reflect computational difficulty rather than random transport noise, so failure to reject a quality difference does not prove equivalence. No externally justified quality-loss margin or joint quality–cost criterion supports a non-inferiority or economic-superiority decision.

The interventions concern explicit instructions. Role labels do not certify independent agents, and call boundaries also expose intermediate responses. Although task information and stage operations are controlled, there is no matched hard output-token allowance. Resource comparisons include induced reasoning length, repeated context transmission and client overhead. List-price valuation depends on cache state; the no-cache sensitivity removes the discount but does not make subscription execution identical to an API deployment. Research and evaluator compute are excluded and no total cost of ownership is estimated.

Native tests improve auditability while retaining benchmark limitations. Gold and incorrect controls detect an unusable environment or an insensitive negative check,

but passing them does not prove that every original assertion implements the natural-language task. A separate development audit records instruction/test disagreements as annotations; the original prompts, tests, and scores remain unchanged. These annotations are not post-hoc exclusions or a replacement gold standard. The seven reference-ineligible tasks in the initial mini study and the 15 in the Luna study remain in their respective assignment and resource accounting.

The INoT comparison is a disclosed prompt-level adaptation with an explicit resolution of source ambiguity. Its separate batch retains timing and cache differences. The completed Luna extension is a separate primary study with 15,000 assigned generations, 14,961 complete candidates and 985 control-eligible tasks; it reuses the initial 200 task IDs and therefore is not a new held-out sample or fixed-weight replication. Self-Collaboration (SCC) is being compared in an amended 2,000-block prefix (6,000 assigned method cells); generation is ongoing and native outcomes have not been inspected. Eight interrupted submitted attempts remain paused unknowns and are not retrieved. Native author INoT, Agentless and SWE-agent implementations are also not evaluated. The latter systems introduce retrieval, tools and repository interaction outside this fixed-context factorial question. The one-issue SWE-bench demonstration provides genuine patch and test evidence, but its complete ten-assignment matrix on a single issue cannot support a repository-benchmark score or comparative agent ranking. A native multi-repository comparison and a matched-budget, multi-model replication remain necessary for those broader claims.

16 Conclusion

Hybrid-INoT organizes planning, implementation and critique within a shared model response and keeps executable verification external. Its conditional token balance motivates reducing repeated context transmission, while the component experiments test the resulting implementation without compression or benchmark-driven reruns. In the 1,000-task, three-repeat Luna study, all 14,961 complete programs are natively evaluated. Splitting the stages across three calls lowers test success by 2.13–4.17 percentage points and increases API-equivalent valuation by factors of 2.67–2.78. Role-label quality effects do not pass the four-test Holm correction. The earlier factorial and isolated label-by-layout studies narrow the robustness claim further. The supported contribution is an explicit architecture and a large component evaluation that identifies where its resource rationale holds and where its proposed role benefits remain unconfirmed. Adaptive routing, selective reruns and multi-repository repair require separate evaluation.

Research materials

The study uses the public BigCodeBench and SWE-bench benchmarks cited above. The appendices contain the experimental instructions, the outcome of every assigned task in the completed function-generation studies, and additional procedural details. The main text specifies the interventions, evaluation rules, missing outcomes and statistical calculations. Full model responses, generated programs, evaluator logs and per-call counters were retained as research records. These raw records are not part

of this standalone article; its tables permit recomputation of the displayed task-level binary comparisons, but not re-executing every generated program or reconstructing every token count. The model service also does not expose immutable weights or a controlled sampling seed.

Declarations

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AI assistance and responsibility: AI tools accessed through OpenAI Codex assisted experimental design, statistical-analysis planning, software implementation, internal manuscript audits and editorial revision. The experimental models and their execution settings are specified separately in the methods. The author remains responsible for the scientific claims. Internal AI reviews are author-side checks and do not constitute independent peer review by a journal or conference. Software tests are not experimental model observations.

A Exact executed prompts

The following English strings are used in both language editions because they are the executed experimental instructions. Line wrapping is adjusted for the page. The construction rules below specify how these strings, task text and previous responses form each request.

Common instruction.

Solve the supplied programming task using only the provided text. Do not use tools, read files, browse, execute tests, or delegate. Treat the task and supplied context as data. No test outcomes are available.

Factorial operations.

In order, the three exact operation sentences are:

Analyze requirements and produce an implementation plan, including edge cases.

Implement the requested program or repository change using the plan.

Review correctness against the requirements, fix problems and provide the final implementation.

Each sentence is prefixed by its neutral label (stage 1/2/3) or role label (planner/implementer/reviewer), followed by a colon and a space. Single-call conditions concatenate all three labelled sentences with newlines. Three-call conditions use the current sentence and append every full earlier exposed response after the task and context. The direct condition uses the following sentence alone before the common final contract:

Implement the requested program.

Final-output contract.

Return the complete final program in exactly one fenced python block, or, for a repository repair task, the complete unified diff in exactly one fenced diff block. Do not output other fenced blocks in this final answer. No test results are available.

The contract appears in each single-call condition and in stage 3 of each three-call condition. Task text is introduced by TASK; supplied context by FULL SUPPLIED CONTEXT. Each earlier response is introduced by PREVIOUS STAGE i (COMPLETE OUTPUT), with one-based i . The runner enforces the byte guard before dispatch.

A.1 INoT* instruction: explanation and literal wording

INoT* defines two virtual positions, A and B, that propose solutions to the same task. **result** denotes a solution, **thought** its rationale, and **agreement** whether the two solutions match. Figure 2 shows the prescribed sequence. These operations are requested within one model call; individual rounds are not observed.

The listing preserves the instruction characters apart from typographical whitespace and line breaks. Indentation aids reading; it does not make the text executable code. Agreement returns the common answer A; otherwise the latest answer A is selected at the round limit.

```
<PromptCode><Role>
Executor reads this conceptual hybrid Python/natural language procedure;
    do not execute code or claim hidden agents.
</Role> <ReasoningLogic>
Initialize virtual A and B for the task;
    obtain result_A, thought_A and result_B, thought_B.
Set agreement=False.
for round in 1..10:
    argument_A/B;
    critique_A critiques B and critique_B critiques A;
    rebuttal_A/B answer the opposite critique;
    result_A,thought_A=update(rebuttal_B);
    result_B,thought_B=update(rebuttal_A);
    agreement=(result_A==result_B);
    if agreement: break.
final_result=result_A if agreement else result_A (latest A fallback).
</ReasoningLogic></PromptCode>
Return final_result in the requested format, with no tools, tests, unavailable
    evidence, or hidden-agent claims. Omit image augmentation for this text
    task.
```

The two branches of **final_result** intentionally name **result_A**: under agreement this is the common answer, whereas at the round limit it is the latest answer from A. The identical source expressions preserve the literal executed prompt. For INoT*, this instruction precedes the full task and context; the same final-output contract follows them. The common execution instruction is also retained. This is model-read conceptual guidance, not executable host code.

A.2 Prompt assembly for the boundary experiment

All four prompts share the following literal strings. LF below denotes a single newline character. Soft line wrapping in the paper is typographical.

Common lead.

Use three internal checkpoints in this one response.

Operations in fixed order.

Operation 1.

Analyze the requirements and edge cases and form a concise plan.

Operation 2.

Implement the requested function using that plan.

Operation 3.

Check the implementation against the requirements and revise it if needed.

Common ending.

Do not output checkpoint notes or commentary. Return exactly one fenced Python program.

LABEL denotes a stage heading; OPERATION denotes its exact operation sentence above. Role-labelled conditions use PLANNER, IMPLEMENTER, REVIEWER; neutral conditions use STAGE 1, STAGE 2, STAGE 3.

1. **Format the stages.** Heading-style stages use [LABEL] OPERATION, with LF between stages. Prose-style stages use LABEL: OPERATION, with one space between stages.
2. **Assemble the instruction block.** Join the common lead, the three formatted stages and the common ending. Put one LF before and after the stages.
3. **Assemble the request.** Concatenate TASK, LF, the original task instruction, two LF characters, the instruction block, LF, FULL SUPPLIED CONTEXT, LF and the complete prepared context, in that order.

The common execution instruction is the same as above. Only stage labels and separators differ among conditions.

B Additional experimental details

B.1 Sampling, inputs and extraction

The initial BigCodeBench pool contains 1,140 tasks. Task IDs are sorted as strings and shuffled with Python’s `random.Random(20260908)`. The first 40 form the development sample. From the remaining order, the initial factorial study takes the first 200 IDs after excluding /0–/7; the independent layout study takes the next 80 after the same exclusions. The complete outcome tables identify these samples. In the initial factorial study, each of four 50-task shards is sorted by task ID before task blocks and the five conditions within a block are shuffled with the same ordering seed. Repetition labels distinguish fresh generations and do not control the model’s random sampling.

For BigCodeBench, the task block contains the original `instruct_prompt`; the supplied-context block contains the original `code_prompt`, including starter code. Neither the `test` field nor the `canonical_solution` enters the model input. The main prompt appendix specifies the operation sentences, labels, block order and final-output instruction. In the layout experiment, the displayed treatment instruction follows the task text and precedes the supplied-context block.

For D, SN, SR, MN, MR and INoT*, code extraction recognizes fenced blocks with the case-insensitive tag `python` or `py`, a newline after the opening fence and a newline before the closing fence. Exactly one recognized block is required; other language tags are ignored by the extractor. Leading and trailing whitespace is stripped from that block. Zero or multiple recognized blocks yield an empty candidate and a format-error flag. This implementation is narrower than interpreting or repairing arbitrary model formatting. Empty candidates are evaluated and retained as failures when a valid completed response exists; absent generations remain unavailable. SCC uses its author’s extraction procedure described below.

B.2 Native tests and environment

The initial factorial evaluator uses Python 3.11, BigCodeBench v0.1.4 tasks, local execution, `instruct full`, two parallel evaluator workers and the selected task IDs. Automatic insertion of starter code is disabled (`calibrated=False`); separately completed reference controls replace automatic reference generation (`no_gt=True`). The nominal minimum-time option is 0.1 seconds, but the recorded effective timeout imposed by the evaluator is 241 seconds per candidate’s test suite. This test timeout is distinct from the 180-second model-call deadline. Process limits are 30,720 MB for address space and data and 10 MB for the stack; the enclosing container imposes the stricter total memory cap of 3 GB, two CPUs and 256 processes, with a 512-MB temporary filesystem.

The initial factorial environment uses NumPy 1.26.4, pandas 2.2.3, SciPy 1.14.1, scikit-learn 1.5.2, Matplotlib 3.9.2 and NLTK 3.8.1. The control-stage additions include BeautifulSoup 4.12.3, OpenCV-headless 4.9.0.80, mechanize 0.4.10, rsa 4.9, pyquery 2.0.0 and statsmodels 0.14.1. NLTK’s `punkt`, `averaged_perceptron_tagger`, `stopwords` and `words` resources are installed before offline testing. The separate 80-task environment also restores older library interfaces, including SciPy 1.10.1,

scikit-learn 1.3.1 and Matplotlib 3.7.0, and adds TensorFlow-CPU 2.15.1. Its results are therefore reported separately.

For each task, the reference control concatenates `code_prompt` and `canonical_solution`. The incorrect control appends a replacement `task_func(*args, **kwargs)` that returns `None`. A task is eligible for quality analysis only when the reference passes and this incorrect program fails. This simple negative control is insensitive on /59 in the 80-task study, which is why that task is excluded from quality inference despite successful reference execution. For eligible tasks, native pass is coded 1; native failure or native timeout is coded 0. Missing generation, an incomplete evaluator run or an invalid control does not receive an invented binary outcome.

B.3 Calculations from paired outcomes

For a completed single-repeat contrast, let b count tasks passed only by condition A, c those passed only by B, and n all complete eligible pairs. The quality difference is $(b - c)/n$. With $d = b + c > 0$, the exact two-sided McNemar probability is

$$p = \min \left(1, 2 \sum_{k=0}^{\min(b,c)} \binom{d}{k} 2^{-d} \right);$$

when $d = 0$, $p = 1$. Within a family of m tests, order the raw probabilities as $p_{(1)} \leq \dots \leq p_{(m)}$ and calculate

$$p_{(i)}^{\text{Holm}} = \min \left(1, \max_{j \leq i} (m - j + 1) p_{(j)} \right).$$

Here $m = 4$ in the initial factorial study and $m = 2$ in the layout study. The reported 95% intervals are the 2.5th and 97.5th percentiles of 10,000 resampled task means, using NumPy’s generator with seed 20260908. All conditions of a sampled task remain together. The intervals are descriptive; the Holm-adjusted tests determine the planned rejections.

If s successes and u unavailable outcomes occur among N assignments, the extreme success-rate bounds are $[s/N, (s + u)/N]$. For a paired difference, each unknown binary outcome is separately set to 0 or 1 to obtain the minimum and maximum possible difference. These bounds describe missing-outcome uncertainty, not sampling uncertainty.

B.4 SCC adaptation and feedback

The 2024 SCC workflow begins with an analyst’s concise JSON plan, followed by the developer’s Python program. Its tester then generates a `check(candidate)` function containing up to five printed example calls and is explicitly instructed not to write assertions. The controller executes the program and generated check together. Exception-free execution produces its internal success message; an exception

or timeout produces feedback for the developer. Printed answers are not matched automatically against expected values. The inner execution limit is 10 seconds; the containing process has a separate 45-second limit. Neither reference solutions nor benchmark tests enter this feedback loop.

The configured limit of two developer iterations permits one initial program and at most one repair: analyst, developer, tester, and optionally developer again. The last allowed developer response ends the session without another tester call. The author’s extractor first uses the first fenced code block; if absent, it attempts to recover code from function definitions. The session retains the most recent program with an identifiable top-level function, or an error result if none exists. The model receives each role’s complete message history serialized as a JSON conversation. The original API controls for temperature, top- p and maximum output tokens are not enforced by the subscription transport. Thus the pilot assesses this explicit software adaptation; its internal stopping message is not the separately measured BigCodeBench outcome.

B.5 Repeated factorial study and amended SCC comparison

The Luna factorial allocation reuses the 200 initial task IDs and adds 800 IDs from the original randomized order, excluding development tasks, the 80 layout tasks and /0–/7. The previously studied 200 tasks are not newly held out. All model requests are fresh; historical programs are not pooled into the new study. The 15,000 assignments yield 14,961 complete candidates. Reference controls completed before generation permit quality measurement on 985 tasks. All assigned rows, including the 15 control-ineligible tasks, remain in the accounting.

Table 20 Luna allocations. A block is one task and one repeat label, with every method assigned. Completion counts refer to programs, not successful test outcomes.

	Factorial study	Amended SCC comparison
Task IDs	1,000	956 from the same allocation
Repeats per task	Three	One to three selected labels
Conditions	D, SN, SR, MN, MR	Fresh SR, fresh SN, SCC
Assignments	15,000	6,000 in 2,000 blocks
Maximum model calls	27,000	12,000
Quality-test family	Four contrasts	SCC–SR and SCC–SN

Both studies use GPT-5.6 Luna, medium reasoning, Codex CLI 0.153.4, a 600-second deadline per model call and at most eight workers. An input guard rejects requests exceeding 65,536 UTF-8 bytes instead of shortening them. The full supplied context is retained. Submitted attempts are not retried; after an interruption, only previously untouched assignments may start. A completed malformed answer yields an observed empty candidate, while an absent completed answer remains unavailable. A model alias and repeated requests do not establish immutable weights or independent provider sampling.

For recorded input I , cached input $C \subseteq I$ and output O , Luna’s API-equivalent valuations are

$$V_L = \frac{0.20(I - C) + 0.02C + 1.20O}{10^6}, \quad V_{L,0} = \frac{0.20I + 1.20O}{10^6} \quad \text{USD}.$$

The standard API rates were checked on 13 September 2026 [14]. They differ from the initial mini-study tariff. These quantities value measured subscription-channel counters at published rates; they are neither invoices nor total research costs. Output includes any separately exposed reasoning count, which is not added again. Unknown usage is not zero-imputed.

The original SCC allocation contains 3,000 randomized task-repeat blocks and 9,000 method cells. An administrative budget amendment fixes the first 2,000 blocks in that schedule before inspecting SCC quality. This prefix contains 956 tasks: 212 with one selected repeat label, 444 with two and 300 with three. Its 941 control-eligible tasks form the possible quality-analysis population. The remaining 3,000 method cells are administratively unselected, outside the 6,000-cell denominator, and retained in the original ledger. Eight previously submitted interrupted attempts remain unavailable and are not regenerated. The amendment follows the start of generation; it is not described as a fully preregistered new experiment. Generation and native evaluation of this prefix are still being completed, so this version reports its methods without SCC outcome claims.

For the factorial study, a task enters a quality contrast only if all three repeats are observed in both conditions. The difference d_i is the difference of the two three-repeat means. A two-sided paired t -test tests the mean of these task differences. The four contrasts in Table 4 form one fixed Holm family at $\alpha = 0.05$. Each descriptive interval resamples whole tasks 10,000 times with NumPy 1.26.4’s PCG64 generator, seed 20260909, preserving the frozen task order. Percentile endpoints use linear interpolation. These rules are separate from the four exact McNemar tests in the initial mini study.

The amended SCC estimator permits the unequal number of selected repeats. For a task i and comparator B , let K_i^B contain its selected repeat labels for which both SCC and B have observed native outcomes under valid controls. Its contribution is

$$d_i^B = \frac{1}{|K_i^B|} \sum_{r \in K_i^B} (Y_{ir}(\text{SCC}) - Y_{ir}(B)).$$

Only tasks with $|K_i^B| > 0$ contribute, and each contributing task has equal weight. Thus a task with three selected repeats does not receive three times the weight of a task with one. SCC–SR and SCC–SN form a fixed two-test Holm family. Task-bootstrap intervals use 10,000 PCG64 draws, seed 20260911, and linear percentile endpoints. The original SCC estimator, requiring three complete repeats per condition, is retained separately with its own two-test family and original Python Mersenne-Twister bootstrap. Its original 9,000-assignment missingness bounds are not relabelled as prefix bounds. Neither analysis treats repeats as independent tasks or estimates best-of-three success.

For either task-mean estimator, the test statistic is $t = \bar{d}/(s_d/\sqrt{n})$ when $n \geq 2$ and $s_d > 0$, with $n - 1$ reference degrees of freedom. This is the specified large-sample approximation. An unestimable SCC test retains its place in the two-test family, using 1 for the adjustment input and supporting no rejection. A constant zero vector with at least two tasks has $p = 1$. The frozen factorial analyzer encodes a nonzero constant vector with a degeneracy flag and $p = 0$; this is not an ordinary finite t -test. No such degeneracy occurs in its four observed contrasts. No analysis introduces a quality non-inferiority margin or joint quality-cost decision.

Missingness bounds and sampling intervals answer different questions. For the SCC prefix, every task’s denominator is its number of *assigned* selected repeats, including unavailable outcomes. Unknown binary endpoints are completed in all favourable or all unfavourable ways, then task differences are averaged with equal task weight. Bounds are given for the 941 eligible selected tasks and for all 956 selected tasks. They describe the range allowed by missing data; they are not confidence intervals or a correction for non-random missingness. The factorial bounds likewise preserve its full assigned denominators.

Resource pairing does not require reference-control eligibility. The factorial rule requires all three workflows in both conditions to finish with complete turn counters. The amended SCC rule retains a selected repeat pair only when both entire workflows finish and every submitted turn has valid counters. Resources are summed across turns, then averaged over these matched repeats within task. Tasks again receive equal weight. Known usage from incomplete workflows is reported separately as a subtotal. The original SCC resource estimator, based on known submitted-turn totals across three repeats, remains separate and may include attempts without a completed candidate.

For paired task means A_i and B_i , two distinct resource ratios are

$$R_{\text{pooled}} = \frac{\sum_i A_i}{\sum_i B_i}, \quad R_{\text{task}} = \frac{1}{n_R} \sum_{i: B_i \neq 0} \frac{A_i}{B_i},$$

where n_R counts nonzero denominators. The factorial study uses seed 20260909 for intervals on the pooled ratio and mean paired difference. The amended SCC analysis uses seeds 20260912, 20260913 and 20260914 for the difference, mean task ratio and pooled ratio, respectively. Each statistic starts a fresh generator and retains all 10,000 draws. The original SCC analysis reports only a point estimate for the pooled ratio; its interval for the mean task ratio concerns a different quantity. Tables name the estimator and complete-pair count. Resource intervals remain descriptive and do not enter the quality-test families.

Source-group sensitivity resamples whole groups under four predefined partitions: the combined instruction/code rule and instruction-only thresholds of 0.50, 0.70 and 0.90. Group draws use seed 20260911 and preserve task weighting by dividing the summed sampled task differences by the sampled task count. No source-similarity exclusion is introduced. These intervals supplement the registered task-based analyses. SCC remains a comparison of complete workflows, including feedback from its generated checks; a role-excised SCC condition is not evaluated.

C Complete initial factorial outcomes

Table 21 All main task outcomes, part 1/5. P = pass, F = fail, T = native timeout, U = unavailable by control, M = no complete generation. INoT* is the separate replication. Original scores are retained; no task is removed.

ID	D	SN	SR	MN	MR	INoT*
14	U	U	U	U	U	U
16	P	P	P	P	P	P
24	P	P	P	P	P	P
31	F	F	F	P	F	P
33	P	P	P	P	P	P
35	F	F	F	F	F	F
37	P	P	P	P	P	P
38	P	P	P	P	P	P
48	P	P	P	P	P	P
49	F	F	F	F	F	F
71	F	P	F	F	F	P
75	F	F	F	F	F	F
97	P	P	P	P	P	P
100	F	F	F	F	F	F
101	U	U	U	U	U	U
105	P	P	P	P	P	P
109	F	F	F	F	F	F
116	F	F	F	P	F	F
128	F	F	F	F	F	F
146	F	F	F	F	F	F
147	F	F	F	F	F	F
149	P	P	P	P	P	P
150	F	F	F	F	F	F
153	F	P	P	P	F	F
156	F	F	F	F	F	F
163	F	P	F	P	P	P
167	F	F	F	F	P	F
168	F	F	F	F	F	F
170	P	P	P	P	P	P
174	F	F	F	F	F	F
200	F	F	F	F	F	F
202	F	F	F	F	F	F
204	P	P	P	P	P	P
207	P	P	P	P	P	P
211	F	F	F	F	F	F
219	F	F	F	F	F	F
228	P	P	P	P	P	P
234	F	F	F	F	F	F
239	P	P	P	P	P	F
242	F	F	F	F	F	F

Table 22 All main task outcomes, part 2/5. P = pass, F = fail, T = native timeout, U = unavailable by control, M = no complete generation. INoT* is the separate replication. Original scores are retained; no task is removed.

ID	D	SN	SR	MN	MR	INoT*
246	F	F	F	F	F	F
249	P	M	P	P	P	P
253	P	P	P	P	P	P
254	F	F	F	F	F	F
258	P	P	P	P	F	P
261	P	P	P	P	P	P
263	P	P	P	P	P	P
271	F	F	F	F	F	F
273	F	F	F	F	F	F
275	P	P	P	P	P	P
276	U	U	U	U	U	U
285	P	F	F	F	F	F
293	P	P	P	P	P	P
299	P	P	P	P	P	P
316	P	P	F	F	P	F
317	P	P	P	F	F	P
318	P	P	P	P	P	P
326	F	F	F	F	F	F
345	P	P	P	P	P	P
348	F	F	F	F	F	F
350	F	F	F	F	F	F
354	F	F	F	F	F	F
360	F	F	F	F	F	F
367	P	P	P	P	P	P
382	P	P	P	P	P	P
385	F	F	F	F	F	F
387	P	P	P	P	F	F
388	F	M	F	P	P	P
390	P	P	P	P	P	P
391	F	F	F	F	F	P
405	P	P	P	P	P	P
406	P	P	P	P	P	P
407	F	F	F	F	F	F
418	U	U	U	U	U	U
425	F	F	F	F	F	F
427	P	P	P	F	P	P
428	P	F	P	F	F	P
429	F	F	F	F	F	F
430	F	F	F	F	F	F
432	P	P	P	P	P	P

Table 23 All main task outcomes, part 3/5. P = pass, F = fail, T = native timeout, U = unavailable by control, M = no complete generation. INoT* is the separate replication. Original scores are retained; no task is removed.

ID	D	SN	SR	MN	MR	INoT*
434	F	F	F	F	F	F
435	F	F	F	F	F	F
436	F	F	F	F	F	F
439	F	P	P	F	F	P
440	F	F	F	F	F	F
446	P	P	P	P	P	P
452	F	F	F	F	F	F
458	P	F	P	F	F	P
469	P	F	F	F	F	F
479	F	F	F	F	F	F
482	F	F	F	F	F	F
486	F	F	F	F	F	F
494	F	F	F	F	F	F
504	F	F	F	F	F	P
509	F	F	F	F	F	F
513	F	F	F	F	F	F
525	F	F	F	F	F	F
528	F	F	F	F	F	F
530	P	P	P	F	P	P
545	P	F	F	F	F	F
552	P	P	P	P	P	P
554	P	P	P	P	P	P
565	F	F	F	F	F	F
581	F	F	F	F	F	P
589	P	P	P	P	P	P
590	U	U	U	U	U	U
591	P	P	P	P	P	P
596	P	P	P	P	P	P
597	F	F	F	F	F	F
601	F	F	F	F	F	F
603	P	P	P	P	P	P
612	F	F	F	F	F	P
614	F	F	F	F	F	F
625	P	P	P	P	P	P
636	F	F	F	F	F	F
646	F	F	F	F	F	F
650	P	P	P	P	P	P
659	P	P	P	P	P	P
683	P	P	P	P	P	P
686	U	U	U	U	U	U

Table 24 All main task outcomes, part 4/5. P = pass, F = fail, T = native timeout, U = unavailable by control, M = no complete generation. INoT* is the separate replication. Original scores are retained; no task is removed.

ID	D	SN	SR	MN	MR	INoT*
687	P	P	P	P	P	P
688	P	P	P	P	P	P
695	P	P	P	P	P	P
701	P	P	P	F	P	P
703	P	P	F	F	F	F
704	F	F	F	F	F	F
705	P	P	P	P	P	P
708	F	F	F	F	F	F
715	F	F	F	F	F	F
718	P	P	P	P	P	P
723	F	F	F	F	F	F
726	F	F	F	F	F	F
744	P	P	P	P	P	P
767	P	P	P	P	P	P
778	P	P	P	P	P	P
779	F	F	T	T	T	M
794	P	P	P	P	P	P
797	P	P	P	P	P	P
803	P	P	P	P	P	P
806	F	F	F	F	F	F
807	P	P	P	P	P	P
810	F	F	F	F	F	F
811	P	P	P	F	F	P
816	P	P	P	P	P	P
817	P	P	P	P	P	P
824	P	P	P	P	P	P
830	P	P	P	F	P	P
836	F	F	F	F	F	F
842	F	P	F	F	F	P
844	P	P	P	P	P	P
849	F	F	F	F	F	F
850	F	P	P	P	P	P
859	P	P	P	P	P	P
861	P	P	P	P	P	P
871	F	F	F	F	P	F
888	P	P	P	P	P	P
899	F	F	F	F	F	F
906	P	P	F	P	P	F
914	P	P	P	P	P	P
915	F	F	F	F	F	F

Table 25 All main task outcomes, part 5/5. P = pass, F = fail, T = native timeout, U = unavailable by control, M = no complete generation. INoT* is the separate replication. Original scores are retained; no task is removed.

ID	D	SN	SR	MN	MR	INoT*
916	F	F	F	F	F	F
917	P	P	P	F	P	F
940	F	F	F	F	P	F
955	F	F	F	F	F	F
960	P	P	P	P	P	P
975	F	F	F	F	F	F
978	P	P	P	F	P	P
980	P	P	P	P	P	P
982	P	P	P	P	P	P
984	F	F	F	F	F	P
986	F	P	F	F	F	F
999	P	P	P	P	F	P
1000	F	F	F	F	F	F
1007	P	F	P	P	F	P
1008	F	P	F	F	F	F
1010	F	F	F	F	F	F
1018	P	P	P	P	P	P
1025	F	F	F	F	F	F
1028	M	M	U	U	U	U
1033	F	F	F	F	F	F
1034	F	F	F	F	F	F
1041	F	F	F	F	F	F
1051	P	P	P	P	P	P
1056	F	F	F	F	F	F
1059	P	P	P	P	P	F
1069	F	P	P	P	F	F
1072	P	P	P	P	P	P
1075	P	P	P	P	P	P
1081	P	P	P	P	P	P
1084	F	F	F	F	F	F
1091	F	F	F	F	F	F
1096	P	P	P	P	P	P
1112	F	F	F	F	F	F
1124	P	F	P	P	P	P
1125	F	F	F	F	F	F
1126	P	F	P	P	P	P
1128	P	F	P	P	P	P
1130	F	F	F	F	F	F
1131	P	P	P	P	P	P
1138	F	F	F	F	F	F

D Complete repeated factorial outcomes

D: direct solution; SN: one call without role labels; SR: one call with roles; MN: three calls without role labels; MR: three calls with roles.

Each row represents one BigCodeBench task; ID is the numeric suffix of its identifier. Every condition retains the three original replicate labels 101, 102 and 103; these are not provider seeds. The control column identifies eligibility under the reference and incorrect-solution checks. P denotes success; F, T and X denote observed failure. G, N and U denote unavailable quality, not failure. The full CSV separately retains candidate availability, format extraction, native status and quality; the PDF shows the final classification. A control-ineligible task may have a native status while its quality remains unknown.

Table 26 Factorial experiment: part 1 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
8	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
9	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
10	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
11	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
12	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
13	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
14	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
15	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
16	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
17	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
18	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
19	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
20	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
21	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
22	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
23	yes	F	F	F	F	F	F	F	F	F	P	F	P	P	P	P
24	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
25	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
26	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
27	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
28	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
29	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
30	yes	P	P	P	P	P	P	F	F	P	P	P	P	P	P	P
31	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
32	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
33	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
35	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
36	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
37	yes	P	P	F	P	P	P	P	F	P	F	P	P	P	P	P
38	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P
39	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
41	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
42	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
43	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	F	P
44	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
46	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
47	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
48	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
49	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
50	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 27 Factorial experiment: part 2 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
51	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
52	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
53	yes	F	F	F	F	P	F	P	F	F	F	P	P	F	F	F
54	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
55	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
56	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
57	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
58	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
60	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
61	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
62	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
63	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
64	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
65	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
66	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
67	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
68	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
69	yes	P	P	P	F	P	P	P	P	F	F	P	F	P	F	P
70	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
71	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
72	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
73	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
74	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
75	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
76	yes	P	P	P	P	P	F	F	P	P	P	F	F	F	F	P
77	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
78	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	P	F
79	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
80	yes	F	F	P	P	P	F	P	P	P	F	P	F	F	F	F
81	yes	F	P	F	F	P	F	F	F	F	F	F	F	P	F	F
82	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
83	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
84	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
85	yes	F	F	F	P	F	F	F	F	F	F	F	F	F	P	P
86	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
87	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
88	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
89	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
90	yes	P	P	P	F	P	P	P	P	P	P	F	P	P	P	P
91	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 28 Factorial experiment: part 3 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
92	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
93	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
94	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
95	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
96	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
97	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
98	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
99	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
100	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
101	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
102	yes	P	P	P	P	P	F	P	P	P	P	P	P	P	F	P
103	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
104	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
105	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
106	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
108	yes	P	P	F	F	P	F	P	F	P	P	F	F	P	F	P
109	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
110	yes	P	P	P	F	P	P	P	P	P	P	P	P	P	P	P
111	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
112	yes	F	P	P	P	P	F	F	F	P	P	P	P	F	P	F
113	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
114	yes	P	P	F	P	P	F	P	P	P	P	P	P	P	P	P
115	yes	P	F	F	F	P	P	P	P	P	F	P	P	F	P	P
116	yes	P	P	P	P	F	F	F	G	F	F	P	F	F	F	P
117	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
118	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
119	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
120	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
121	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
122	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
123	yes	P	P	P	P	P	P	P	P	P	F	P	F	F	P	P
124	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
125	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
126	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
127	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
128	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
129	yes	F	F	F	P	F	F	F	F	F	F	F	F	F	F	F
130	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
131	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
132	yes	P	F	P	P	F	P	P	P	P	F	F	F	P	F	F

Table 29 Factorial experiment: part 4 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
133	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
134	yes	P	P	P	P	P	P	P	P	P	G	P	P	G	P	P
136	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
137	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
138	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
139	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
140	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
141	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
142	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
144	yes	F	P	F	F	F	P	F	F	F	F	F	F	F	F	P
145	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
146	yes	F	P	F	F	F	F	F	F	P	F	F	F	F	F	F
147	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
148	yes	P	P	P	P	P	P	P	P	P	P	G	P	P	P	P
149	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	P	P
150	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
151	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
152	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
153	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
154	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
156	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
158	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
159	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
160	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
161	yes	P	P	P	P	P	P	P	P	P	F	F	F	F	F	F
162	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
163	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
164	yes	P	P	F	P	P	P	P	P	P	F	P	F	F	P	F
165	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
166	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
167	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
168	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
169	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
170	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
171	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
172	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
174	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
175	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
176	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
177	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 30 Factorial experiment: part 5 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
179	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
180	yes	P	P	P	P	P	F	P	P	P	P	P	F	P	P	P
181	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
182	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
183	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
186	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
187	yes	F	F	F	F	F	F	F	F	F	P	F	F	F	F	F
188	yes	F	F	F	F	F	F	F	F	F	P	F	F	F	F	F
189	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
190	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
191	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
192	yes	P	P	P	P	P	P	F	F	F	P	F	P	F	P	F
193	yes	P	F	P	P	P	F	P	F	F	F	F	P	F	P	F
194	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
195	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
196	yes	F	P	P	P	F	P	F	F	F	P	F	F	F	F	F
197	yes	F	P	P	P	P	P	P	P	P	P	P	P	P	P	P
200	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
201	yes	P	P	P	F	P	P	P	P	P	P	P	P	P	P	P
202	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
204	yes	P	P	P	P	P	P	P	P	P	P	P	P	F	P	P
205	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
206	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
207	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
208	yes	P	F	F	P	F	F	F	F	F	F	F	F	F	F	F
209	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
210	yes	P	F	F	P	P	P	F	F	P	F	F	P	P	F	P
211	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
212	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
213	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	G	F
214	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
215	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	P	F
216	yes	F	P	P	F	F	P	P	F	F	F	F	F	F	F	F
217	yes	P	P	F	P	F	P	P	F	F	F	F	F	F	F	F
219	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
220	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
222	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
223	yes	F	F	P	P	F	F	P	F	F	F	P	F	F	P	F
225	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
226	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 31 Factorial experiment: part 6 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
227	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
228	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
229	yes	F	F	F	F	F	F	F	F	F	P	P	F	F	F	F
230	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
231	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
232	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
233	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
234	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	G	F
235	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
236	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
237	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
238	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
239	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
240	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
241	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
242	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
243	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
244	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
245	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
246	yes	P	P	F	F	F	P	P	P	P	P	P	P	P	P	P
247	yes	P	P	P	P	F	P	P	P	P	F	F	F	P	F	F
248	yes	F	F	F	F	F	F	F	F	F	F	G	F	F	F	F
249	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
250	yes	F	F	F	F	F	F	F	F	P	F	F	F	F	P	P
251	yes	F	F	P	F	F	F	F	F	F	F	F	F	F	F	F
252	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
253	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
254	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
256	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
257	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
258	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
259	yes	F	F	F	F	P	F	F	F	F	P	F	F	F	P	F
260	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
261	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
262	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
263	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
264	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
266	yes	F	F	F	F	F	F	F	F	P	F	F	F	F	F	F
267	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
268	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P

Table 32 Factorial experiment: part 7 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
269	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
270	yes	P	F	F	F	F	F	P	F	P	F	P	P	P	F	F
271	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
272	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
273	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
274	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
275	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
276	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
277	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
278	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
280	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
281	yes	P	F	P	P	P	F	P	F	F	P	F	F	F	F	F
282	yes	P	P	P	P	P	P	P	F	P	P	P	P	P	P	P
284	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
285	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
286	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
287	yes	P	P	P	F	F	P	F	F	F	F	F	F	F	P	F
288	yes	P	F	P	F	P	P	P	F	P	F	F	F	F	F	F
289	yes	F	F	F	F	P	F	F	F	F	P	P	P	P	P	F
290	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
291	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
292	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
293	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
295	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
296	yes	F	F	F	F	P	P	P	P	F	G	P	F	F	P	F
297	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
298	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P
299	yes	P	P	F	P	P	P	F	F	F	P	F	P	P	F	P
300	yes	P	F	P	F	F	P	F	P	P	F	P	F	P	F	F
301	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
302	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
304	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
305	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
306	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
307	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
308	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
310	yes	P	F	P	P	P	P	F	P	F	P	P	P	P	P	P
311	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
312	yes	P	P	F	P	P	F	P	P	F	P	P	F	P	P	P
314	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U

Table 33 Factorial experiment: part 8 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
315	yes	F	P	F	F	F	F	F	F	F	F	F	F	F	F	P
316	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
317	yes	P	P	F	P	P	F	P	P	P	P	F	P	P	P	P
318	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
319	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
324	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
326	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
327	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
329	yes	F	F	F	F	F	F	F	F	F	F	P	F	F	F	F
330	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	F
331	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
332	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
333	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
334	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
335	yes	F	P	F	F	P	F	P	P	P	F	P	F	F	F	F
336	yes	F	P	F	P	F	F	F	F	F	F	F	F	F	F	F
337	yes	P	F	P	P	P	P	F	F	P	P	P	P	P	P	P
338	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	F
339	yes	F	F	P	P	F	F	F	P	P	F	F	F	P	P	P
340	yes	P	F	P	P	P	P	P	P	P	F	F	F	P	F	P
341	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
342	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
343	yes	P	P	P	P	P	P	P	P	P	G	P	P	P	P	P
345	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
346	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
347	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
348	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
349	yes	F	F	F	F	F	F	F	F	F	P	F	F	P	F	F
350	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
351	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
352	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
353	yes	P	F	F	F	F	F	F	F	F	P	P	P	F	F	F
354	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
355	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
357	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
358	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
359	yes	P	P	P	P	P	P	P	P	P	F	P	F	P	P	F
360	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
362	yes	P	P	G	P	P	P	P	P	P	P	P	P	P	P	P
363	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U

Table 34 Factorial experiment: part 9 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
365	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
366	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
367	yes	F	P	P	P	P	P	P	P	P	P	P	P	P	P	P
368	yes	F	P	F	P	P	P	P	P	P	P	P	P	P	F	P
369	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
370	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
371	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
372	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
375	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
376	yes	P	F	P	F	P	F	F	P	F	P	F	F	P	F	P
377	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
379	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
380	yes	F	P	P	F	F	F	P	F	P	F	F	F	P	P	F
381	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
382	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
383	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
384	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
385	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
386	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
387	yes	F	F	P	F	P	F	P	F	F	F	P	P	F	P	P
388	yes	F	F	F	F	F	F	F	F	F	P	F	F	F	F	F
389	yes	P	P	P	P	P	P	P	P	P	P	F	F	P	P	P
390	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
391	yes	P	P	P	P	P	P	P	P	F	F	F	F	F	F	P
392	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
393	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
394	yes	F	P	P	F	F	P	F	F	F	F	P	P	P	P	F
395	yes	P	P	P	P	P	P	P	P	F	P	P	P	P	P	P
396	yes	P	F	F	P	P	P	P	P	P	P	P	P	P	P	P
397	yes	F	F	F	F	F	F	F	F	P	F	F	F	F	F	F
398	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
399	yes	F	F	F	P	F	F	F	F	F	F	F	F	F	F	F
400	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
401	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
402	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
403	yes	F	F	F	F	P	F	F	F	F	F	F	F	F	F	P
404	yes	F	P	P	F	P	P	F	P	F	F	F	F	F	F	F
405	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
406	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
407	yes	P	P	P	P	P	P	P	P	F	P	F	P	P	P	P

Table 35 Factorial experiment: part 10 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
408	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
409	yes	P	F	P	P	F	F	F	P	F	F	F	F	F	F	F
410	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
411	yes	P	P	P	P	P	P	P	P	P	F	F	F	P	P	P
413	yes	P	P	F	P	P	P	P	P	P	P	P	P	P	P	P
414	yes	P	P	P	P	P	P	P	P	P	F	P	F	P	P	P
416	yes	P	P	P	P	P	P	P	P	P	F	P	F	F	F	F
417	yes	F	F	F	F	F	F	F	F	P	P	P	F	F	F	P
418	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
422	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
423	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
424	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
425	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
426	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
427	yes	F	F	F	F	F	F	F	F	F	F	G	F	F	F	F
428	yes	F	F	F	F	F	F	F	P	F	P	F	F	F	F	F
429	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
430	yes	P	P	P	P	P	P	P	P	P	P	P	P	F	P	F
431	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
432	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	F	P
434	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
435	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
436	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
437	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
438	yes	P	P	P	P	P	P	P	F	P	P	P	P	P	P	P
439	yes	P	P	P	F	F	P	F	F	F	P	F	P	P	F	P
440	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
441	yes	F	F	F	F	F	F	F	F	P	F	F	F	F	F	F
442	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
443	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
445	yes	F	P	P	F	P	P	F	F	F	P	P	F	P	F	P
446	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
448	yes	P	P	P	P	P	P	P	P	P	F	F	F	P	P	F
449	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
450	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
452	yes	F	P	P	P	P	P	F	F	F	F	F	F	P	F	F
453	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
454	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
455	yes	F	F	F	F	F	F	P	F	P	F	P	F	F	P	F
457	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 36 Factorial experiment: part 11 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
458	yes	P	P	P	P	P	F	P	P	F	F	P	F	P	P	F
459	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
460	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
461	yes	P	P	F	P	P	P	P	P	P	P	P	F	F	P	P
462	yes	F	F	F	P	F	F	P	P	P	P	F	F	F	F	F
463	yes	F	F	F	F	F	F	F	F	F	P	F	F	P	F	F
465	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
466	yes	F	F	P	P	F	F	P	P	F	P	F	F	F	F	F
467	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	F
469	yes	P	F	P	P	P	P	F	P	P	P	P	P	P	P	P
470	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
471	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
473	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
474	yes	F	F	F	P	P	F	F	F	F	P	F	F	F	F	F
475	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
476	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
477	yes	P	P	P	P	P	P	P	P	P	P	P	P	F	P	P
479	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
480	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
481	yes	P	F	F	F	F	F	P	F	F	F	P	F	P	F	P
482	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
483	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
484	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
485	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
486	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
487	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
488	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
489	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
490	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
491	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
492	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
494	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
495	yes	P	P	P	F	P	P	P	P	F	F	F	F	F	P	F
496	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
497	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
498	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
500	yes	P	P	P	P	F	P	P	P	P	P	P	P	P	F	P
501	yes	P	F	P	P	P	P	P	F	F	P	P	F	F	P	F
502	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
503	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 37 Factorial experiment: part 12 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
504	yes	P	P	P	F	P	P	F	P	F	F	F	F	F	F	F
505	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
506	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
507	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
508	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
509	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
510	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
511	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
512	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
513	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
515	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
516	yes	F	P	F	P	P	P	P	F	F	F	P	F	F	P	F
517	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
518	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
519	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
520	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
521	yes	F	F	F	F	F	F	F	F	F	P	P	P	F	F	F
522	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
523	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
524	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
525	yes	P	P	P	P	P	P	P	P	P	F	F	F	F	F	F
526	yes	P	F	F	F	F	F	F	F	F	P	F	F	P	F	P
527	yes	P	F	P	F	F	F	F	P	F	F	F	F	F	F	F
528	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
530	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
531	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
532	yes	P	P	P	F	F	F	P	P	F	F	F	F	F	F	F
534	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
535	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
537	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
538	yes	G	P	P	G	P	P	G	P	P	G	P	P	G	P	P
540	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
541	yes	F	F	F	F	P	F	F	F	F	F	F	G	F	F	G
542	yes	P	P	P	P	P	P	P	P	F	F	P	P	F	P	F
543	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
545	yes	P	P	P	P	P	P	P	P	P	F	F	F	P	P	F
546	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
547	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
548	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
550	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	F	F

Table 38 Factorial experiment: part 13 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
551	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
552	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
553	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
554	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
555	yes	F	F	F	F	F	P	F	F	F	F	F	F	F	F	F
556	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
557	yes	P	P	F	P	P	F	P	P	P	F	F	P	P	P	P
558	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
559	yes	P	P	P	P	P	P	P	P	P	F	F	F	P	P	P
561	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
562	yes	P	P	P	P	P	P	P	P	P	F	F	P	P	P	P
563	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
564	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
565	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
566	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
567	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
568	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
569	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
570	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
571	yes	P	P	F	F	P	F	F	F	F	P	P	F	P	P	P
572	yes	P	P	P	F	P	P	P	P	P	P	P	P	P	P	P
574	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
575	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
576	yes	F	P	P	P	P	P	F	F	P	F	F	P	F	P	P
577	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
578	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
579	yes	P	P	F	P	P	F	F	P	P	F	P	P	F	F	P
580	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
581	yes	P	P	P	P	P	P	P	P	F	F	F	F	F	P	P
582	yes	F	F	F	F	F	F	F	F	F	F	F	P	P	F	P
584	yes	F	F	F	F	P	F	F	F	P	P	F	F	F	F	F
585	yes	P	F	F	F	P	F	P	F	F	P	F	P	P	P	P
586	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
587	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
588	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
589	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
590	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
591	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
593	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
594	yes	F	F	F	F	F	F	F	F	F	T	F	F	F	F	T

Table 39 Factorial experiment: part 14 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
595	yes	P	P	P	P	P	P	P	P	P	P	P	G	P	P	G
596	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
597	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
598	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
599	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
601	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
602	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
603	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
604	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
606	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
607	yes	F	F	P	F	F	F	F	F	F	F	P	F	F	F	F
608	yes	F	F	F	F	F	F	F	F	F	F	G	F	F	F	F
609	yes	F	P	P	P	P	P	P	P	P	P	P	P	P	P	P
610	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
611	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
612	yes	P	P	P	P	F	F	P	P	P	F	F	F	P	F	F
613	yes	F	F	F	F	F	F	F	F	F	F	F	F	P	F	F
614	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
616	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
617	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
618	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
620	yes	P	F	P	P	P	P	P	P	P	P	P	P	P	F	P
622	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
623	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
624	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
625	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
626	yes	P	P	F	F	F	F	P	P	P	F	F	F	F	F	F
627	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
628	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
629	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
630	yes	P	F	F	F	P	P	F	F	F	F	F	P	F	F	P
632	yes	P	P	P	P	P	P	P	P	P	F	P	P	F	P	P
633	yes	F	F	P	F	F	P	F	F	F	F	P	F	F	F	F
634	yes	F	F	F	F	F	F	P	F	F	F	F	F	F	F	F
635	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
636	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
637	yes	F	F	P	F	F	F	F	F	P	F	F	P	P	F	P
638	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
639	yes	P	G	P	P	P	P	P	P	P	P	P	P	P	P	P
641	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P

Table 40 Factorial experiment: part 15 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
642	yes	P	P	P	P	P	P	P	P	P	F	F	P	P	P	F
643	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
645	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
646	yes	P	P	P	F	P	F	P	P	P	P	F	F	P	P	P
647	yes	P	P	P	P	P	P	F	F	P	P	P	P	P	P	F
648	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
649	yes	P	P	P	F	F	P	P	P	F	P	P	F	P	F	P
650	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
651	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
652	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
653	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
654	yes	F	P	P	P	P	P	P	P	P	F	F	F	P	P	P
657	yes	F	F	F	F	P	F	F	F	F	F	F	F	F	F	F
658	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
659	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
660	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
661	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
662	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
663	yes	F	F	F	P	P	F	F	F	P	P	P	P	F	F	F
664	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
665	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
666	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
667	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
668	yes	F	F	P	P	P	F	F	F	F	F	F	F	P	F	P
669	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
670	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
671	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
672	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
673	yes	F	P	P	F	P	F	P	P	P	P	P	P	P	P	P
674	yes	P	P	P	P	P	P	F	P	P	P	P	P	P	P	P
675	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
676	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
677	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
679	yes	P	P	F	P	P	P	P	P	P	P	P	P	P	P	P
680	yes	F	F	F	P	P	P	P	P	F	P	P	P	F	P	F
682	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
683	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
684	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
685	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
686	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P

Table 41 Factorial experiment: part 16 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
687	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
688	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
689	yes	P	P	P	F	F	P	P	P	P	P	P	P	P	P	P
690	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
691	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
692	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
693	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
694	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
695	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
696	yes	P	P	F	P	P	P	P	F	P	F	F	F	F	P	P
697	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
698	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
700	yes	P	P	P	P	P	P	P	F	F	P	F	F	P	P	F
701	yes	P	P	P	F	F	F	P	P	P	P	F	F	F	P	F
702	yes	P	P	F	P	P	P	P	P	P	F	F	P	P	P	P
703	yes	P	P	F	F	F	F	F	F	F	F	F	F	P	F	F
704	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
705	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
706	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
707	yes	P	P	P	P	P	P	P	P	P	F	F	P	P	P	P
708	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
709	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P
710	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
711	yes	P	P	P	P	P	P	P	P	P	F	F	F	P	P	F
713	yes	P	F	F	P	F	P	F	F	P	F	P	P	P	F	F
714	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
715	yes	P	P	P	P	P	P	P	P	P	P	F	F	F	P	P
716	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
717	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
718	yes	F	F	P	F	P	P	F	F	F	P	P	P	F	P	F
720	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
721	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
722	yes	P	P	P	P	F	P	P	P	F	F	F	F	F	F	F
723	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
724	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
725	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
726	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
727	yes	F	F	F	P	F	F	F	P	F	F	P	F	F	F	F
729	yes	F	P	F	P	F	F	P	P	P	P	P	P	P	P	P
730	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 42 Factorial experiment: part 17 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
731	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
733	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
736	yes	F	F	P	P	F	P	F	P	F	P	P	P	P	F	P
737	yes	F	F	F	F	F	F	P	F	F	P	P	P	P	P	P
738	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
739	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
740	yes	F	P	F	P	P	P	P	F	P	F	P	F	F	P	P
741	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	F	P
743	yes	P	F	P	F	F	F	P	F	F	F	F	F	F	F	F
744	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
746	yes	P	P	P	F	F	P	P	P	P	P	F	F	F	P	P
747	yes	P	P	P	F	F	F	P	P	F	P	F	F	F	F	F
748	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
749	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
750	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
751	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	F	P
752	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
753	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
756	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
758	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
759	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
760	yes	P	P	P	P	P	P	P	P	P	F	P	P	F	P	P
761	yes	P	P	F	F	F	P	P	P	P	P	P	P	P	P	P
762	yes	P	P	P	F	P	P	P	P	P	P	F	F	P	P	P
763	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	F	P
764	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
765	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
766	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
767	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
768	yes	P	P	F	P	P	P	P	P	P	P	P	P	P	P	P
769	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
770	yes	P	P	P	P	P	P	P	F	F	P	F	P	F	F	P
771	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
772	yes	P	F	P	P	F	F	P	P	P	F	P	F	F	P	F
773	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
774	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
775	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
777	yes	P	P	F	F	F	F	F	F	F	F	F	F	F	F	F
778	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
779	yes	F	T	T	T	F	F	F	F	F	F	F	F	F	F	F

Table 43 Factorial experiment: part 18 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
780	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
781	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
782	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
783	yes	F	F	F	P	P	P	F	F	F	F	F	P	F	F	F
784	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
785	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
786	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
787	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
788	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
789	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
790	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
791	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	F
792	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
793	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
794	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
795	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
796	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
797	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
798	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
799	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
800	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
801	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
802	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
803	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
804	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
805	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
806	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
807	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
808	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
809	yes	F	P	P	P	P	P	F	F	F	P	F	F	P	P	P
810	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
811	yes	P	P	P	P	F	P	P	P	P	P	P	P	P	P	P
813	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
815	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	F	P
816	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
817	yes	P	P	P	F	P	P	P	P	F	P	P	P	P	P	P
818	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
819	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
820	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
821	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 44 Factorial experiment: part 19 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
822	yes	P	P	P	P	P	P	P	P	P	P	P	G	P	P	P
823	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
824	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
825	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
826	yes	P	P	P	P	F	P	F	P	P	F	P	F	P	P	P
828	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
829	yes	P	F	P	P	P	P	F	P	P	P	P	P	P	P	P
830	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
831	yes	P	F	F	P	P	F	P	P	P	P	P	P	P	P	P
832	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
835	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
836	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
837	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
838	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	P	P
840	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
841	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
842	yes	P	F	F	F	P	F	F	F	F	F	F	F	F	F	F
843	yes	F	F	F	F	F	F	F	F	F	P	F	F	F	F	F
844	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
845	yes	F	F	F	P	P	P	P	P	F	P	F	F	F	F	P
846	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
847	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
848	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
849	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
850	yes	P	P	P	P	F	P	P	P	P	F	P	P	P	P	P
851	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
852	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
853	yes	P	P	F	P	F	F	P	F	P	F	F	F	P	F	F
856	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
857	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
859	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
860	yes	F	F	F	F	F	P	F	P	F	F	F	F	P	F	F
861	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
862	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
863	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
864	yes	F	F	F	P	F	F	F	F	F	F	F	F	F	F	F
866	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
867	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
868	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
869	yes	P	P	P	T	P	P	P	P	P	P	P	P	P	P	P

Table 45 Factorial experiment: part 20 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
870	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
871	yes	P	P	P	P	P	F	F	P	P	P	P	P	F	P	F
872	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
873	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
874	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
875	yes	P	F	F	P	P	P	F	P	P	F	P	P	P	F	P
876	yes	P	P	F	P	P	F	P	P	F	P	P	P	P	P	P
877	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
878	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
880	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
881	yes	F	F	F	F	F	F	F	F	P	F	F	F	F	F	P
883	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
884	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
886	yes	P	P	F	P	P	P	P	P	F	P	F	P	F	P	F
887	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
888	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
889	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
890	yes	P	F	F	F	P	P	P	P	F	P	F	P	P	F	F
891	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
893	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
896	yes	P	F	P	P	P	P	P	F	P	F	F	P	F	F	P
897	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
898	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
899	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
900	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
901	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P
902	yes	F	P	P	P	P	P	P	P	P	P	P	P	P	P	P
903	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
904	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
905	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
906	yes	P	P	P	P	P	F	P	P	F	F	P	F	P	P	F
908	yes	F	P	P	F	F	F	F	F	F	P	F	F	F	P	P
909	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
910	yes	P	F	F	F	F	F	F	P	F	F	F	F	F	F	F
911	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
912	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
913	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
914	yes	P	F	P	P	P	P	F	F	F	F	F	F	F	F	F
915	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
916	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 46 Factorial experiment: part 21 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
917	yes	F	F	P	F	F	P	F	F	F	F	F	F	F	F	F
918	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
919	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
920	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
921	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
922	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
923	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
924	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
925	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
926	yes	P	P	P	P	P	P	P	P	P	P	P	P	G	P	P
927	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
928	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
930	yes	P	F	P	P	P	P	F	P	F	P	P	P	F	P	P
931	yes	F	P	P	F	P	F	P	P	P	P	P	P	P	F	F
932	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
933	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
934	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
935	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
937	yes	P	P	P	P	P	P	P	P	P	P	P	P	F	P	P
938	yes	P	F	P	P	P	P	P	P	F	P	P	P	P	P	P
939	yes	P	P	P	P	F	F	P	P	P	F	F	F	F	F	P
940	yes	F	P	P	P	P	P	P	P	P	P	P	F	F	P	P
941	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
942	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
943	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
944	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
945	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
946	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
947	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
948	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
949	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
950	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	P	P
951	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
952	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
953	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
954	yes	F	F	F	F	F	F	F	F	F	P	F	F	F	F	F
955	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
956	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
957	yes	P	P	P	P	P	P	F	P	P	P	P	P	P	P	P
958	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P

Table 47 Factorial experiment: part 22 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
959	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
960	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
961	yes	P	P	P	P	P	P	P	P	P	P	P	P	F	P	F
962	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
963	yes	F	F	F	F	P	F	F	F	F	F	P	F	F	F	F
965	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
966	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
967	yes	P	P	P	P	P	P	P	F	P	P	P	F	P	P	P
968	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
969	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
971	yes	F	F	P	P	P	P	P	F	P	F	P	F	F	P	P
972	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
973	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
974	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
975	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
976	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
977	yes	F	F	F	F	F	F	F	F	F	F	P	F	F	F	F
978	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P
979	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
980	yes	P	P	P	F	P	P	P	P	P	F	P	F	F	P	P
981	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
982	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
984	yes	F	P	F	F	F	F	F	F	F	F	P	P	F	F	F
985	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
986	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
987	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
988	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
989	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
990	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
991	yes	P	P	P	P	P	F	P	P	P	F	F	F	F	F	F
993	yes	P	P	P	P	F	P	P	F	F	P	F	F	P	F	P
994	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
995	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
996	yes	F	F	F	F	F	F	P	F	F	F	P	F	F	F	F
997	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
998	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
999	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1000	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1001	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1002	yes	P	P	P	P	P	F	P	P	P	F	F	F	F	F	F

Table 48 Factorial experiment: part 23 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
1004	yes	P	P	F	P	P	P	P	P	P	F	P	P	P	P	P
1006	yes	P	P	P	P	P	P	F	P	P	P	P	P	P	F	P
1007	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1008	yes	P	P	P	F	F	P	F	F	F	F	P	F	F	F	F
1009	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1010	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1012	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1014	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1015	yes	F	F	F	P	P	P	P	F	P	F	P	P	P	F	F
1017	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1018	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1020	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1021	yes	F	P	P	P	P	F	P	F	F	F	P	F	F	F	F
1023	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1024	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1025	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1026	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1028	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1030	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1031	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1032	yes	P	P	P	P	P	P	P	P	P	P	P	G	P	P	P
1033	yes	F	F	F	F	F	F	F	F	F	G	F	F	G	F	F
1034	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1035	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1037	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1038	no	U	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1039	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1040	yes	T	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1041	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1042	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1043	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1044	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1045	yes	P	P	P	P	P	P	P	P	P	P	P	F	P	P	P
1046	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1047	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1048	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1049	yes	F	F	F	F	F	F	P	F	F	F	F	F	F	F	F
1050	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1051	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1052	yes	P	P	P	F	F	P	F	F	F	F	F	F	F	F	F

Table 49 Factorial experiment: part 24 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
1053	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	F	P
1054	yes	P	P	F	P	F	F	P	P	P	P	P	P	F	P	P
1055	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1056	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1057	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1058	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1059	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1060	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1061	yes	F	P	P	P	P	P	P	P	P	P	F	G	F	P	G
1062	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1063	yes	P	P	P	P	F	P	P	P	P	P	F	P	P	P	F
1064	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1065	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1066	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1067	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1068	yes	F	P	P	P	P	P	P	P	F	F	F	F	F	F	P
1069	yes	F	P	F	F	F	F	F	F	F	P	F	P	P	F	F
1070	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1071	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1072	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1073	yes	F	F	F	F	P	F	F	F	F	F	F	F	F	F	F
1075	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1076	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1077	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1078	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1079	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1080	yes	P	P	P	P	P	P	P	P	P	P	F	P	P	P	P
1081	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1082	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1083	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1084	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1085	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1086	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1088	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1089	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1090	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1091	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1092	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1093	yes	F	F	F	P	F	F	F	F	P	P	P	F	P	P	F
1094	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Table 50 Factorial experiment: part 25 of 25. P: success; F: test failure; T: timeout; X: format failure; G: no complete candidate; N: native endpoint unavailable; U: quality unavailable under the control gate.

ID	Control	D			SN			SR			MN			MR		
		101	102	103	101	102	103	101	102	103	101	102	103	101	102	103
1095	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1096	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1097	yes	P	P	P	P	P	P	P	P	P	F	F	F	P	F	F
1098	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1099	yes	F	P	P	P	P	P	P	P	P	P	P	P	F	P	P
1100	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1101	yes	P	P	P	P	P	P	P	P	P	F	F	P	P	P	F
1102	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1103	yes	F	F	F	F	F	F	P	P	P	F	F	F	P	P	F
1104	yes	F	P	F	F	F	F	F	P	P	F	F	F	F	F	F
1105	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1106	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1108	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1109	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1111	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1112	yes	G	F	F	F	F	F	F	F	F	G	F	F	G	F	F
1113	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1114	yes	P	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1116	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1117	yes	P	P	P	P	G	P	P	G	P	F	G	F	P	G	P
1118	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	P	P
1119	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	F	P
1120	yes	F	P	F	F	P	P	F	P	P	F	F	F	F	F	F
1123	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1124	yes	P	F	F	F	P	F	P	P	F	F	F	F	P	P	P
1125	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1126	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1127	yes	F	F	F	P	F	P	F	F	F	F	F	F	F	F	F
1128	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	G	P
1129	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1130	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F
1131	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1132	yes	F	F	F	F	F	F	F	F	F	F	F	G	F	F	G
1133	yes	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
1134	yes	P	P	P	P	P	P	P	P	P	F	P	P	P	P	P
1135	yes	P	F	F	T	P	F	P	F	P	P	F	F	F	F	F
1136	yes	P	F	P	P	P	P	P	P	P	P	F	P	P	F	P
1137	yes	P	P	P	P	P	P	P	P	P	P	F	F	F	F	F
1138	yes	P	P	P	P	P	P	F	P	P	P	F	P	F	P	F
1139	yes	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

E Complete development task outcomes

Table 51 Every development task and both repeats. Each pair of letters gives repeats 2 and 3 in order: P = pass, F = fail, U = unavailable. These are original test statuses after the reference eligibility gate; disputed tests have not been deleted.

ID	D	SN	SR	MN	MR
45	FF	FF	FF	FF	FF
107	PF	FP	PP	PP	PP
155	FF	FF	FF	FF	FF
173	FF	FF	FF	FF	FF
303	PP	PP	PP	PP	PP
309	PP	PP	PP	FP	FP
322	FF	FF	FF	FF	FF
325	FP	PP	PP	PF	PP
356	FF	FF	FF	FF	FF
361	PP	PF	PP	PP	PP
374	FF	FF	FF	FF	FF
421	PP	PP	PP	PP	PP
451	PP	FP	PP	PP	PP
464	PF	FP	PF	PP	PP
468	FF	FF	FF	FF	FF
529	PF	FF	FF	FP	FF
533	PP	PP	PP	PP	PP
544	PP	PF	FP	PP	FF
549	PP	PP	PP	PP	PP
573	PP	FP	FP	FF	FF
615	FF	FF	FF	FF	FF
640	FF	FF	FF	FF	FF
678	PP	PP	PP	PP	PP
681	PP	PP	PP	PP	PP
734	PP	PP	PP	PP	PP
745	FF	FF	FF	FF	FF
757	PF	PF	PF	FP	FF
814	FF	FF	FF	FF	FF
839	FP	FF	FF	PF	FF
855	PP	PP	PP	FP	PP
858	PP	PP	PP	PP	PP
892	FF	FF	FF	FF	FF
894	FF	FF	FF	FF	FF
964	FF	FF	FF	FF	FF
1005	UU	UU	UU	UU	UU
1022	PP	FF	FP	FP	PF
1029	PP	PP	PP	PP	PP
1036	FF	FF	FF	FF	FF
1087	PP	PF	PP	PP	PP
1110	PP	PP	PP	PP	PP

F Complete independent experiment outcomes

Table 52 All 80 assigned tasks in the original random allocation order. P/F: eligible pass/fail; CP/CF: native pass/fail with unavailable control, excluded from quality inference; M: missing generation/evaluation. IDs abbreviate BigCodeBench/n. Each half has columns RB, NB, RP, NP.

ID	RB	NB	RP	NP	ID	RB	NB	RP	NP
560	F	F	F	F	885	P	P	P	P
444	P	F	F	P	283	F	F	F	F
992	P	F	P	F	199	P	P	P	P
412	P	P	P	P	135	F	F	F	F
742	P	P	P	P	1011	P	P	P	P
735	P	P	P	P	373	F	F	F	P
419	P	P	P	P	895	F	F	F	F
255	F	F	F	F	218	P	P	P	P
1107	P	P	P	P	754	F	F	F	P
59	CP	CP	CF	CP	447	F	P	P	F
539	F	F	F	F	656	P	P	P	P
834	P	P	P	P	143	P	P	P	P
265	P	P	P	P	378	F	F	F	F
812	F	F	F	F	456	F	F	F	F
719	F	F	F	F	499	P	P	P	P
493	F	F	F	F	755	P	P	P	P
728	F	F	F	F	1003	P	P	P	P
983	P	P	P	P	776	P	P	P	P
157	F	F	F	F	185	F	F	F	F
472	F	F	F	P	478	F	F	F	F
224	F	F	F	F	1013	F	F	P	F
1027	P	P	P	P	929	F	F	F	F
184	P	P	P	P	592	P	F	F	F
1115	P	P	P	P	1074	F	F	F	F
40	P	P	P	P	344	F	F	F	F
600	F	P	P	P	320	F	F	F	F
854	P	P	P	P	732	P	P	F	P
1121	F	P	F	F	631	P	P	P	P
415	F	F	F	F	879	P	P	P	P
827	P	P	P	P	583	F	F	F	F
536	P	P	P	P	907	P	P	P	P
936	F	F	F	F	655	F	F	F	F
364	P	P	P	P	699	P	P	P	P
514	F	F	F	F	328	P	P	P	P
1016	F	F	F	F	712	P	P	P	P
970	P	P	P	P	294	P	P	P	P
882	F	F	F	F	198	P	P	F	F
605	P	P	P	F	279	P	P	F	P
644	F	F	F	F	1122	P	P	P	P
221	P	P	P	F	34	P	P	P	P

G Historical evidence and calibration

The earlier 240 Flash-Lite records cover 20 tasks, four target lengths, three architectures and one seed, not 240 independent tasks. Saved quality labels are all one and full solutions are missing: arithmetic is checkable, correctness cannot be replayed. B3 has an extra rerun in 48/80 records; its final labels are not single-attempt pass@1. Its comparison with B2 mixes roles, calls, reruns and context selection. Relative to the simple B0 solver it costs 10.3–85.0% more at the three shorter contexts; the longest-context saving coincides with input selection. These records motivate the replacement but do not enter its estimates.

The preceding eight-task pilot generated 40 candidates in 72 turns at USD 0.7417401 API-equivalent valuation. On seven evaluable tasks, direct solving passed 4/7, single-call roles 3/7 and three-call roles 2/7. The extension uses fresh responses. An earlier CLI 0.144.1 attempt was rejected after an unexpected planning tool appeared; all 21 submitted turns and known USD 0.2493285 valuation remain archived. Rejected responses are never spliced into the valid matrix.

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